2025-2026

Supervised by Dr.Heba El-Shimy

RESEARCH REPORT

EEG–EMG Fusion for Upper-Limb Movement Recognition Using Machine Learning and Deep Learning Models

Project by: Muskaan Garg

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GitHub Link: https://github.com/Muskaan2058/eeg_emg_fusion_ml

Submitted in partial fulfilment for the degree of BSc Hons Computer Science (Artificial Intelligence).

Heriot-Watt University

Computer Science

School of Mathematical and Computer Sciences

Student Declaration of Authorship

Course code and name:	F20PA – Research Methods and Requirements Engineering
Type of assessment:	**Individual**
Coursework Title:	Honour project - Dissertation (D3)
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Student ID Number:	H00416442
Programme:	BSc Computer Science
Dissertation / Project Title:	EEG–EMG Fusion for Upper-Limb Movement Recognition Using Machine Learning and Deep Learning Models
Supervisor:	Heba El-Shimy

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I focused on improving the performance of the models through a series of refinements - in particular improving the EMG pipeline, diagnostic analysis and fusion architecture design - to maximize the quality of the results and evidence of evaluation presented in the dissertation as much as possible.

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Acknowledgements

I would like to express my deepest gratitude to my supervisor for her invaluable guidance, useful comments and continuous support throughout the development of this project. Her vision has greatly influenced the direction of my work and ensured methodological clarity.

I also acknowledge the F20DL teaching team for the academic foundation, structured resources and technical access that made this work possible. Their support throughout the module has contributed significantly to my understanding of the various concepts and techniques used for my project.

In addition, I extend my appreciation to the university for providing an environment facilitative to independent research and study. Easy access to computing resources, academic literature, and a supportive learning infrastructure has played a vital role in completing this project.

Abstract

This project investigates a multimodal method for classifying upper-limb movements by combining electroencephalography (EEG) and electromyography (EMG) signals. EEG measures brain activity signals prior to movement intention, while EMG records muscle activity during execution. Many studies often analyze these signals separately, which limits recognition accuracy and timing details.

Machine Learning (ML) refers to computational methods that learn patterns from data and use trained models to make predictions automatically. In this study, machine learning models, including multimodal, are used to analyse relationships within EEG and EMG signals and classify gestures.

This study implements and evaluates five deep learning models - EEGNet, EEG CNN-LSTM, EMGNet, EMG CNN-LSTM, and a multimodal fusion model - on the NeBULA dataset (Garro et al., 2025), a 128-channel EEG and 11-muscle EMG dataset of upper-limb reaching tasks across 36 subjects. A subject-independent evaluation protocol is used throughout, with all models evaluated on held-out test subjects unseen during training. Additionally, a temporal pipeline is built to explore the timing difference between EEG and EMG signals.

Three research topics are addressed in this study: which architecture best captures movement-discriminative patterns in each modality, whether fusion outperforms single-modality performance, and what temporal relationship exists between cortical and muscle activity during reaching. Performance is evaluated using classification accuracy and macro F1-score.

1. Introduction

1.1 Background & Motivation

Electroencephalography (EEG) and electromyography (EMG) are two complementary bio-signals that reflect the brain–muscle interaction during movement. EEG measures electrical potentials generated by cortical neurons through scalp electrodes, providing millisecond-level insight into neural activity before motion initiation (Schomer and Silva, 2018). EMG, in contrast, records the electrical activity produced by muscles during movement. It reflects the peripheral activation of the muscle fibres during movement. Together, EEG and EMG provide the basis for tracking movement-related signals in clinical, rehabilitation and research settings to help diagnose neurologic disorders such as epilepsy, Parkinson's disease or other neuromuscular disorders along with designing assistive technologies for motor restoration (Nicolas-Alonso and Gomez-Gil, 2012).

Despite their importance, each has its own limitation when analyzed independently. EEG bio-signals are susceptible to noise, nonstationary, and artifacts (e.g., motion, other external sources of interference) that may mask low level neural activity associated with movement intention (Lotte et al., 2018). On the other hand, EMG signals are typically less noise-prone and are strongly correlated with the amount of force applied to a limb. However, EMG signals are only activated after a movement begins and thus it is unable to predict pre-movement intentions (Zhang et al., 2022). Consequently, systems using only EEG or EMG will typically lack either full timing coverage or reliability in real-world conditions.

Recent advances in machine learning (ML) and deep learning (DL) have enabled researchers to better capture the complexity and nonlinearity inherent in neural–muscular patterns. Recurrent architectures (e.g. LSTM) and convolutional architectures (e.g. CNN) have demonstrated success in modelling spatial and temporal patterns in EEG and EMG time series (Sakhavi, Guan and Yan, 2018; Chiang Liang Kok et al., 2024). Yet, most previous studies have focused on single modality data (either EEG or EMG), and the benefits of multimodal fusion remain largely unexplored.

By integrating EEG and EMG features, fusion frameworks can combine the early predictive power of EEG with the accurate movement signals from EMG, improving gesture classification. This multimodal approach improves accuracy, reduces false detections, and supports natural user interfaces for prosthetics, rehabilitation, and human-computer interaction (Zhang et al., 2022).

1.2 Role of Machine Learning in Bio-signal Analysis

Machine Learning (ML) provides computational methods that learn patterns from data and use trained models to make predictions automatically. Instead of following fixed rules, ML models continue to improve through training on example data and identifying relationships within the data. This helps it make predictions or decisions automatically, even with new unseen data. This makes ML especially useful for EEG and EMG signals, which are often noisy, complex, and difficult to interpret manually.

In EEG–EMG research, ML helps identify patterns in brain activity (EEG) and muscle activity (EMG) that happen during movement. Modern deep learning models, such as CNNs and LSTMs, can automatically extract important features—like frequency patterns or muscle activation changes—without the need for hand-crafted feature design. This leads to better accuracy and more reliable predictions.

Because of these capabilities, ML is used in many applications of EEG-EMG, such as rehabilitation, prosthetic control and brain-computer interfaces (BCIs), which require fast and reliable recognition of human movement intention.

1.3 Research Problem

Although EEG and EMG are each commonly used for gesture recognition and motor intention, a vast majority of past research has evaluated them as individual modalities (Nicolas-Alonso and Gomez-Gil, 2012; Lotte et al., 2018). Despite their complementarity, not many multimodal frameworks thoroughly examine how combining these signals can boost performance of gesture detection using publicly available datasets.

Previous research on the individual signal types has shown that they come with their own limitations. EEG based approaches measure pre-motor cortical activity but suffer from lower signal-to-noise ratios, while EMG-based systems excel at detecting muscle activation but lag in capturing neural intent (Nicolas-Alonso and Gomez-Gil, 2012; Lotte et al., 2018). Moreover, the literature has no consensus regarding the amount of time or proportion of early neural activity versus late muscle activity. Furthermore, comparative analyses between spatial convolutional models such as EEGNet and EMGNet and temporal sequence architectures like CNN–LSTM remain scarce.

Therefore, it is necessary to develop systematic benchmarks that evaluate fusion-based deep learning architectures for gesture classification using synchronized EEG-EMG datasets.

1.4 Aim & Objectives

The integration of EEG and EMG signals presents a promising direction to increase the accuracy and the interpretability of the recognition of gestures. This is done through the combination of analysing cortical activity prior to the movement, along with muscular activation during movement, which will allow for faster and better recognition (Zhang et al., 2022; Nicolas-Alonso and Gomez-Gil, 2012).

Deep and machine learning models – specifically, hybrid and convolutional – are well-suited to capture such complex temporal-spatial relationships contained in the multimodal data (Sakhavi, Guan and Yan, 2018; Chiang Liang Kok et al., 2024) , thus enabling this project to combine both EEG and EMG data to eliminate the limitations of each modality alone (Lotte et al., 2018).

Aim:

The aim of this project is to design, implement, and evaluate a deep learning pipeline that fuses EEG and EMG signals for upper-limb movement classification. This will be used in assessing whether spatial-temporal models (EEGNet, EMGNet) and sequential models (CNN-LSTM) benefit from multimodal fusion compared to single-modality baselines. Additionally, it aims to characterise the temporal relationship between cortical and muscular signals during the reaching tasks.

Objectives:

1. Implement preprocessing pipelines for synchronized EEG and EMG data.
2. Train and evaluate EEGNet, EMGNet and CNN–LSTM models for single-modality and fused inputs.
3. Compare and analyse fusion and non-fusion performance using metrics like accuracy and macro-F1 scores.
4. Analyse the temporal delay between EEG motor onset and EMG muscle activation

1.5 Report Outline

The dissertation is organised into seven chapters that develop, implement, and evaluate the EEG-EMG fusion pipeline for upper-limb movement classification.

Chapter 1 - Introduction explains the background and motivation for combining EEG and EMG signals, defines the research problem, and states the aim and objectives of the project.

Chapter 2 - Background Research examines the fundamentals of EEG and EMG, reviews single-modality and fusion studies, outlines common preprocessing techniques and deep learning architectures, and identifies the research gaps that motivate this project.

Chapter 3 - Methodology describes and justifies the methodological approach, including the NeBULA dataset, preprocessing pipeline, experimental design across five models, and evaluation strategy. It also formally states the three research questions addressed by the project.

Chapter 4 - Feasibility evaluates the viability of the proposed pipeline through preliminary signal analyses on the NeBULA dataset, confirms the technical software and hardware setup, and addresses ethical and legal considerations regarding data use.

Chapter 5 - Implementation and Contributions presents the complete technical implementation of the pipeline, covering preprocessing, subject-independent split strategy, diagnostic analyses for both modalities, EMG and EEG model development, fusion architecture design, and the timing analysis pipeline.

Chapter 6 - Evaluation evaluates all five models against the three research questions, presents quantitative results for each modality and the fusion model, and provides critical interpretation of performance outcomes through diagnostic and timing analyses.

Chapter 7 - Discussion and Conclusion recontextualises the findings, summarises the project's contributions, reflects on limitations, identifies directions for future work, and provides a final standalone conclusion.

2. Background Research

This chapter reviews existing studies on EEG, EMG, and their fusion for gesture recognition. **Section 2.1** describes the fundamentals of EEG and EMG techniques. **Section 2.2** discusses EEG-based motor decoding approaches, while **Section 2.3** focuses on EMG-based gesture recognition. Section 2.4 reviews EEG–EMG fusion studies. **Sections 2.5 and 2.6** outline common preprocessing techniques and relevant deep-learning architectures, and **Section 2.7** summarises key research gaps motivating this project.

2.1 EEG and EMG Fundamentals

EEG and EMG are the two key bio-signals for studying neural-muscular mechanisms in intentional movement. EEG records brain activity using scalp electrodes, it measures voltage changes across frequency bands: Delta (0.5–4 Hz), Theta (4–8 Hz), Alpha (8–12 Hz), Beta (13–30 Hz), and Gamma (30–100 Hz) (Schomer and Silva, 2018). EMG measures electrical activity from contracting muscle fibres, typically ranging from 10–500 Hz, and offers crucial information on muscle activation and force (De Luca, 2002).

Both modalities have limitations. EEG is sensitive to artefacts like eye blinks, facial movements, and electrical noise, requiring time-consuming pre-processing for removal (Lotte et al., 2018). Although EMG is less affected by external factors, it does not detect neural preparation before movement. It only senses activity after motor commands reaches the muscles (Zhang et al., 2022).

Therefore, combining EEG and EMG will eliminate the individual drawbacks associated with each modality. The early neural cues of motor planning provided by EEG can be complemented by the detailed and accurate muscular activity provided by EMG. Combining the two modalities provides a complementary physiological and temporal signal space that enhances the accuracy of predicting movement (Nicolas-Alonso and Gomez-Gil, 2012; Zhang et al., 2022).

2.2 EEG-based Motor Decoding

EEG-based motor decoding has been the primary focus of brain–computer interface (BCI) research, aiming to convert cortical activity into meaningful representations of motor intention. The traditional BCIs included handcrafted feature extraction methods like Common Spatial Patterns (CSP) coupled with machine learning algorithms like Support Vector Machines (SVM) to classify motor imagery. One such research work had results which achieved classification accuracy rates of 70-80% (Blankertz et al., 2008). However, these systems suffered from issues associated with low signal-to-noise ratios (SNR) and high variability in cortical activity between subjects.

Advancements in deep learning then greatly improved the performance of EEG decoders. Schirrmeister et al. (2017) made significant contributions to this field when they developed Convolutional Neural Networks (CNNs), such as DeepConvNet and ShallowConvNet, both of which can automatically extract spatial and temporal features from unprocessed EEG data, thereby increasing classification accuracy to approximately 85%, and generating F1-scores greater than 0.80 compared to typical machine learning approaches.

Lawhern et al. (2018) subsequently created EEGNet, a compact CNN architecture tailored for use with EEG-based BCIs. It was tested across multiple datasets including PhysioNet, OpenBMI, and BCI Competition IV and was able to achieve classification accuracy of 80-88%, with AUC values ranging from 0.85 to 0.90 while also being computationally efficient.

These studies signify a shift from hand-crafted features to deep learning methods in EEG motor decoding. However, challenges such as poor cross-subject transferability, signal variability, and a lack of physiological interpretation of outputs continue to pose challenges (Lotte et al., 2018). Despite these issues, EEGNet represents a significant advancement in producing highly accurate and generalizable EEG-based motor decoders.

2.3 EMG-based Gesture Recognition

Gesture recognition based on EMG, which performs gesture recognition using electrical signals generated from muscles, is widely applied in prosthetic control and human-computer interaction. Early research by Hudgins et al. (1993) utilised a artificial neural network (ANN) classifier with features extracted in the time-domain to achieve an accuracy of about 90%. The disadvantage of this approach was that it required manual feature extraction for each user.

Like EEG, deep learning techniques consequently replaced traditional hand-crafted pipeline approaches. A convolutional neural network (CNN) was applied to the NinaPro DB5 dataset by Atzori, Cognolato and Müller (2016) to obtain average accuracy rates of 75-85% when classifying multiple gestures. This demonstrated that CNN's can be trained to learn both spatial and temporal muscle pattern activity. Chiang Liang Kok et al. (2024) developed a deep-learning model for proportional myoelectric control and achieved R^2 values of > 0.80 and F1-scores of ~ 0.85, greatly exceeding previous results of regression models for predicting movement.

Typical preprocessing steps within an EMG pipeline include band pass filtering (typically between 10-500 Hz), full wave rectification, segmentation of signals into overlapping window sizes, and normalization to account for differences in amplitude among users (De Luca, 2002).

Despite strong performance, EMG-only systems are subject to several limitations including high variability among individuals, variability associated with electrode displacement or fatigue, and lack of capability to detect pre-movement neural activity (Atzori, Cognolato and Müller, 2016).

2.4 EEG–EMG Fusion Studies

Recent research on multimodal EEG–EMG fusion demonstrates that integrating cortical (EEG) and muscular (EMG) activity can improve gesture recognition, yet most studies differ in scope, datasets, and depth of modelling. Some of these research works are discussed below.

Tryon & Trejos (2021), used a CNN (Convolutional Neural Network) to fuse EEG and EMG spectra at the input level to recognize upper limb gestures. The results were an average accuracy of 80.5 ± 8.1%, as well as an F1-Score of 80.7%. However, the study involved a very low number of subjects and no standardization their subjects or sessions.

Similar results were seen by Zhang et al. (2022). They were able to achieve high accuracy (over 95%) in their offline trials and 87% in their real time trials using a CNN-LSTM model that fused both temporal and spatial EEG/EMG signals to recognize lower limb movements. However, the research studied gait rather than upper limb gestures.

Pritchard, Campelo and Goldingay (2025) proposed a framework for recognizing upper limb gestures that tested early and feature-level fusion strategies. The studies found that subject-independent fusion resulted in a multi-gesture classification accuracy of 73% while subject-independent unimodal EEG resulted in 49%

accuracy. Their Hierarchical fusion strategy resulted in 86% accuracy for subject-specific classifications but lacked sequence learning and cross-subject validation. Although, this clearly demonstrated the effectiveness of fusion methods over unimodal ones.

Sultan et al. (2025) introduced NeuroFusion-Trans, a transformer-based EEG–EMG fusion model that achieved 96–97% accuracy and Cohen's Kappa of 0.95–0.97 across two public datasets. While Li et al.'s model performed well it displayed poor robustness to signal noise and overlapping activations and therefore indicated improvement concerning artifact reduction.

Pritchard (2024) from Aston University developed three multimodal fusion architectures and applied his novel "Hierarchical" fusion strategy within an Automated Machine Learning (CASH) Pipeline. Pritchard's subject independent EMG – EEG fusion model resulted in an accuracy of 73.4% whereas a unimodal EEG accuracy was approximately 51.9%. Despite showcasing the advantages of fusion, his work explored classical machine learning techniques rather than modern Deep Learning techniques. The dataset used was also non-standardized, hindering broader benchmarking comparisons.

Study	**Dataset / Task**	**Fusion Level & Model Type**	**Evaluation Setup**	**Key Results / Metrics**	**Main Limitations**
Tryon & Trejos (2021)	In-house dataset; Upper-limb gesture classification	Input-level fusion; CNN using EEG–EMG spectrograms	Subject-specific (small sample)	Accuracy 80.5 ± 8.1%, F1 = 80.7%	Small, non-standard dataset, limited temporal modelling
Zhang et al. (2022)	Custom lower-limb dataset; movement intention detection	Hybrid CNN–LSTM**;** time-synchronised EEG–EMG fusion	Offline and online evaluation	Offline >95%, Online >87% accuracy	Focus on gait, lacked upper-limb generalisation
Pritchard, Campelo and Goldingay, (2025)	Public upper-limb dataset; multi-gesture classification	Early / Feature / Hierarchical fusion with CNN	Subject-independent & subject-specific	73% (subject-independent fusion), 86% (hierarchical subject-specific)	No sequence learning; limited cross-subject validation
Sultan et al. (2025)	Public EEG–EMG dataset; upper-limb gestures	Attention-guided CNN–BiLSTM	Cross-subject validation	Accuracy 95%, Macro-F1 = 0.91	Sensitive to signal noise and overlapping activations; limited generalisation
Pritchard (2024)	Aston University dataset; upper-limb gestures	Classical ML (SVM, RF); Hierarchical fusion via AutoML (CASH)	Subject-independent & subject-specific	Fusion 73.4%, Unimodal EEG ≈51.9%	Uses classical ML; lacks deep temporal or spatial learning; non-standard dataset

Table 1- Summary of comparative EEG–EMG fusion studies

A summary of key multimodal EEG–EMG fusion studies is presented in Table 1 above, comparing how each approach contributes to understanding the fusion effectiveness and its current research gaps. Overall, while these studies confirm the advantages of EEG–EMG fusion, there remain clear gaps: limited deep-learning studies on upper-limb, open-access datasets, lack of temporal analyses, and the absence of comparative benchmarks between spatial and temporal models. This project addresses these shortcomings by systematically evaluating EEGNet, EMGNet and CNN–LSTM fusion models on a public dataset, evaluating both performance and timing relationships.

2.5 Preprocessing Techniques

Preprocessing is important across most of the EEG-EMG fusion literature, as it improves the quality of the signals through which deep learning models learn the dataset. Most studies use band-pass filters on EEG (0.5 – 50 Hz) to obtain the motor-related rhythms (mu (8 – 12 Hz) and beta (13 – 30 Hz)) while removing slow drifts and high frequency noise (Lawhern et al., 2018). Many researchers have used Independent Component Analysis (ICA), as well as Common Average Referencing (CAR), to eliminate artefacts (eye-blink, line noise, muscle) from both EEG and EMG signals (Delorme and Makeig, 2004).

EEG and EMG signals are then windowed into fixed-length overlapping segments to maintain temporal context while reducing the noise, an action particularly beneficial to hybrid temporal models (CNN-LSTM) (Sakhavi, Guan and Yan, 2018).

In addition to segmenting, many fusion papers utilize either the Short-Time Fourier Transform (STFT) or wavelet-based spectral analysis to convert time domain signals to time-frequency representations (spectrograms) suitable for CNN inputs (Tryon & Trejos, 2021; Pritchard, Campelo and Goldingay, 2025). Finally, z-score normalization or min-max scaling are commonly utilized to standardize signal amplitudes for all trials and participants, improving model convergence and generalization.

2.6 Deep Learning Architectures

The deep learning models used in this project are selected to capture both the spatial distribution and temporal components of EEG and EMG signals. EEGNet is a compact, spatial-temporal Convolutional Neural Network that uses depth-wise and separable convolutions to identify spatial filters from short EEG segments (Lawhern et al., 2018). Similarly, EMGNet is a 1-D Temporal CNN which extracts muscle activation patterns over short periods of time from raw EMG signal wave forms (Atzori, Cognolato and Müller, 2016).

In contrast, the CNN–LSTM hybrid model combines the strengths of convolutional and recurrent layers. CNNs extract local temporal–spatial features, while Long Short-Term Memory (LSTM) layers model temporal dependencies over time (Sakhavi, Guan and Yan, 2018). This architecture is particularly suited for the EEG–EMG fusion, as it can integrate the short-term neural activations with later muscle responses to accurately capture the neural-to-muscular activity.

Together, these models provide enhanced frameworks for multimodal decoding, efficiency, and high performance on physiological time-series data.

2.7 Summary & Research Gap

The reviewed literature indicates significant progress regarding EEG- and EMG-based movement recognition, yet several limitations persist. Research concerning EEG-based models evolved from using handcrafted feature extraction (i.e. CSP + SVM) to deep learning models such as EEGNet and CNN–LSTM but each still faces issues related to non-stationarity and cross-subject variability. (Schirrmeister et al., 2017; Lawhern et al., 2018). Similarly, EMG-based models produce highly accurate results on datasets. However, they are subject-dependent and lack predictive capabilities before movement onset (Chiang Liang Kok et al., 2024; (Atzori, Cognolato and Müller, 2016).

Fusion studies of EEG–EMG models have shown that combining neural and muscular features enhances performance and robustness (Pritchard, Campelo and Goldingay, 2025; Tryon and Trejos, 2021). Despite this, several gaps in research persist. These include gaps in the standardization of multimodal datasets, limited comparison of spatial (EEGNet) and temporal (CNN-LSTM) architecture models, and little investigation of the temporal relationship between cortical and muscular activity. Moreover, previous studies are often based on proprietary datasets or non-reproducible pipelines, thereby limiting their applicability (Sultan et al., 2025; Pritchard, 2024).

This project will address these identified gaps using an open access EEG–EMG dataset for the identification of upper limb movement and the implementation of a systematic and reproducible pipeline to compare unimodal and multi-modal models while utilizing consistent pre-processing and evaluation methods. Furthermore, the goal of this study is to provide new insights into the relationship of neural and muscular complements in movement decoding and bridge the gap between early neural intentions and late muscular execution.

3. Methodology and Work Plan

This chapter describes the methodological approach and experimental design for the project. Section 3.1 presents the three research questions. Section 3.2 describes the NeBULA dataset, including subject exclusions and the decision to use the free condition only. Section 3.3 explains the design rationale for the staged modelling approach. Section 3.4 details the preprocessing pipeline for both EEG and EMG. Section 3.5 describes the experimental design including the five models and evaluation protocol. Section 3.6 outlines the evaluation strategy and metrics. Section 3.7 summarises the overall workflow.

3.1 Research Questions and Goals

This study addresses the key research gaps defined in Chapter 2, including the limited use of EEG-EMG fusions, the lack of comparison regarding spatial (EEGNet), and temporal (CNN-LSTM) models, and the ambiguity surrounding the timing of neural and muscle activity. The research therefore focuses on three core questions:

RQ1. Does EEG-EMG fusion outperform single-modality EEG and EMG models for subject-independent upper-limb movement classification?

RQ2. Which architecture - compact spatial-temporal CNN (EEGNet, EMGNet) or hybrid CNN-LSTM - better captures movement-discriminative patterns in each modality?

RQ3. What is the temporal delay between EEG motor onset and EMG muscle activation during reaching?

These questions are addressed through a fully reproducible pipeline on the NeBULA dataset, enabling fair benchmarking across modalities and architectures using consistent preprocessing and evaluation methods.

3.2 Dataset Description

This study uses the NeBULA dataset (Neural and Body signals for Upper Limb Activities published in Nature Scientific Data. NeBULA provides synchronised high-density EEG (128 channels, 1000 Hz) and surface EMG (11 muscles, 1000 Hz) recordings from 40 able-bodied participants performing three variants of upper-limb reaching tasks under three conditions: unassisted (free), low-resistance, and high-resistance exoskeleton assistance (Garro et al., 2025).

Although NeBULA includes recordings across three conditions, this project uses only the free (unassisted) condition exclusively for all analyses - preprocessing, model training, evaluation, and timing analysis. The free condition represents natural, unassisted reaching without exoskeleton resistance, providing the most ecologically valid baseline for establishing movement classification performance. Analysis of the low and high resistance conditions is left as future work.

Three subjects (Sub-03 and sub-12 and sub-21) were excluded prior to analysis due to documented signal quality issues and incomplete recordings and subject 28 was missing from the NeBULA dataset, leaving 36 usable subjects (Garro et al., 2025).

Each subject completed 10 repetitions per reaching task, so there were 30 unassisted tasks performed by each subject that the project focuses on. Movement onset markers are provided in hardware-synchronised event files, enabling precise epoch alignment across both modalities.

NeBULA was selected over alternatives for several reasons. It provides the highest channel count EEG available for upper-limb reaching research, with 128 channels compared to the 8 channels typical of prior datasets. Both modalities are recorded at matched sampling rates with hardware synchronisation, removing the need for post-hoc alignment. The dataset includes a sufficient number of subjects for subject-independent evaluation, and is openly available under a Creative Commons licence with full BIDS-compliant organisation.

The dataset is available at: https://doi.org/10.6084/m9.figshare.27301629.

3.3 Design Rationale

The methodology aims to distinguish the effects of each modality and architecture before bringing them together. T Two modality-specific model types are evaluated in parallel. The first type employs compact spatial-temporal convolutional architectures—EEGNet (Lawhern et al., 2018) for EEG and a 1D CNN (EMGNet) for EMG. The second type uses CNN-LSTM hybrid architectures, which add a bidirectional LSTM layer and attention pooling to show how activation patterns change over time during a trial (Sakhavi, Guan, and Yan, 2018).

Training single-modality models initially enables the independent characterisation of each architecture's strengths and limitations prior to fusion. A multimodal fusion model then combines the best representations from each modality. This staged approach makes sure that any changes in performance seen in the fusion model are due to the combination of modalities and not to differences in architecture.

All five models share the same subject-independent data split, the same random seed, and identical evaluation metrics, enabling direct and fair comparison across all conditions.

Along with classification, a timing analysis pipeline is used to analyse the temporal relationship between EEG cortical motor preparation and EMG muscle activation. Quantifying this EEG-to-EMG delay provides insight into how the two modalities relate physiologically.

3.4 Data Preprocessing

EEG and EMG signals are pre-processed independently using the free-condition recordings. All preprocessing is done in Python using MNE-Python, NumPy, and SciPy.

For EEG, 15 motor-relevant channels are selected from the 128-channel array, covering the sensorimotor cortex bilaterally (C3, C4, Cz, FC3, FC4, CP3, CP4, C1, C2, C5, C6, FC1, FC2, CP1, CP2). A 4th-order Butterworth bandpass filter (1–45 Hz) is applied to remove slow drifts and high-frequency noise, followed by common average referencing (CAR) to suppress volume-conduction artefacts. Signals are then resampled to 200 Hz and z-score normalised per channel.

For EMG, all 11 recorded muscles are retained. A 4th-order Butterworth bandpass filter (20–400 Hz) removes movement artefacts and DC offsets. Signals are full-wave rectified and smoothed using a 100 ms RMS envelope window to obtain muscle activation envelopes, then resampled to 200 Hz and z-score normalised per channel.

Epochs are extracted from −500ms to +2000ms relative to each movement onset marker, producing trial arrays of shape (n_trials, 15, 500) for EEG and (n_trials, 11, 500) for EMG. Epochs are then segmented using a sliding window of 80 samples and a step of 40 samples, yielding 11 windows per trial.

3.5 Experimental Design

Five models are trained and evaluated under a consistent subject-independent protocol. All training and evaluation uses the free-condition recordings exclusively. The dataset is divided at the subject level into training (25 subjects), validation (5 subjects), and test (7 subjects) sets, approximating a 70/15/15 split.

Modality	Model	Architecture Type	Notes
EEG	EEGNet	Temporal + depthwise spatial Conv2d	Compact CNN using depth wise–separable convolutions to learn frequency-specific spatial filters from EEG signals (Lawhern et al., 2018).
EEG	CNN–LSTM (EEG)	Temporal + spatial Conv2d, BiLSTM	Learns temporal dependencies in EEG data; integrates local spatial features with long-term neural activity patterns (Sakhavi, Guan and Yan, 2018).
EMG	EMGNet	Three-block 1D CNN	Learns temporal – special patterns from EMG waveforms for muscle-based gesture classification (Atzori, Cognolato and Müller, 2016).
EMG	CNN–LSTM (EMG)	1D CNN, BiLSTM	Captures sequential muscle activation dynamics and long-term correlations in EMG time-series.
EEG + EMG	Fusion Model	Feature-Level Multimodal Network	Combines feature representations from the best-performing EEG and EMG branches

Table 2- Model Configurations and Descriptions

The experimental design follows a staged approach which is shown in *Table 2*. Single-modality models are trained first - two for EEG and two for EMG - allowing each modality's performance to be established independently before fusion is attempted. The fusion model is then trained using both modalities, enabling a direct comparison between unimodal and multimodal performance under identical evaluation conditions.

3.6 Evaluation Strategy

All five models are evaluated on the same held-out test subjects using two main metrics: classification accuracy and macro-averaged F1-score. Macro F1 is used as the primary metric because it treats all three classes equally, giving a clearer picture than accuracy alone especially when the class distributions might not be balanced across subjects. Confusion matrices are reported for each model to spot any consistent misclassification patterns across the three reaching tasks.

The subject-independent evaluation protocol ensures that no subject present in training or validation contributes to test performance. This gives a realistic estimate of how well the models generalize to new, unseen users, which is a standard practice in BCI research (Lotte et al., 2018).

In addition to classification performance, a timing analysis is conducted to address RQ3. Event-related desynchronisation (ERD) onset is estimated from the EEG alpha band, and EMG activation onset is estimated using a threshold on the RMS envelope. The EEG-to-EMG delay is then quantified per subject and per task, which provides to characterize the timing sequence from the cortex to the muscles during reaching.

3.7 Workflow Summary

The workflow of this project starts from the initial loading and inspection of the EEG-EMG data set to assure that all signals are properly synchronized, then applying preprocessing techniques to each modality (filtering, segmenting, and aligning). Unimodal models are trained first: EEGNet and CNN–LSTM for EEG, and EMGNet and CNN–LSTM for EMG, allowing spatial–temporal and sequential patterns to be learned. The most predictive EEG and EMG embeddings will be fused create a hybrid multimodal classifier. Additionally, a temporal pipeline will be implemented to analyse the EEG-to-EMG delay. The results will then be evaluated to conclude the aim of the research.

4. Feasibility

This chapter evaluates the feasibility of the proposed pipeline through preliminary analyses, technical assessment, and ethical review. Section 4.1 presents early exploratory analyses on the NeBULA dataset, including EEG channel correlation, power spectral density, EMG envelope, and frequency distribution comparisons across three representative subjects. Section 4.2 outlines the technical feasibility of the software stack and hardware environment. Section 4.3 addresses ethical and legal considerations regarding the dataset. Section 4.4 discusses the project's significance and expected outcomes.

4.1 Early Exploration & Preliminary Work

This section demonstrates the feasibility of the proposed EEG-EMG fusion pipeline through exploratory analyses conducted on the NeBULA dataset. Three representative subjects (01, 02, and 04) from the free condition were selected for visualisation.

All analyses were conducted in Python using MNE-Python, SciPy, NumPy, and Matplotlib, with code available in the project GitHub repository.

4.1.1 EEG Channel Correlation Heatmap

A channel correlation heatmap shows the pairwise Pearson correlation coefficients for all 15 motor channels. It is used to determine whether EEG electrodes capture spatially structured brain activity rather than random noise, which is required for spatial convolution operations in EEGNet (Lawhern et al., 2018).

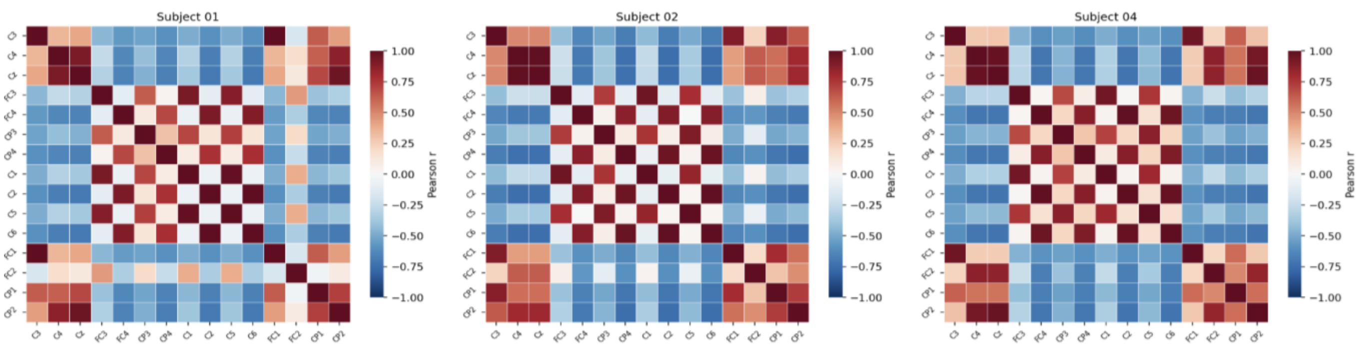

Figure 1- EEG Channel Correlation Heatmap (task free)

Figure 1 shows the correlation heatmaps for Subjects 01, 02, and 04. There is a consistent checkerboard-like pattern across all three subjects. There are strong positive correlations (dark red, r approaching 1.0) between channel pairs that are close together in space and functionally related, and negative correlations (blue) between pairs that are farther apart. Channels in the same hemisphere and functional region, like C3, FC3, and CP3 on the left sensorimotor cortex, group together with high positive correlations. Channels in opposite regions, on the other hand, show negative correlations.

This spatial organization is consistent and reproducible across subjects, indicating that the EEG data contain structured cortical activity rather than random noise. The consistency of this pattern across subjects confirms that the 15-motor channel selection identifies significant spatial dependencies, confirming the dataset's suitability for spatial-temporal deep learning models.4.1.2 EEG Power Spectral Density (PSD) - Mean ± SE by Gesture.

4.1.2 Power Spectral Density (PSD) for EEG

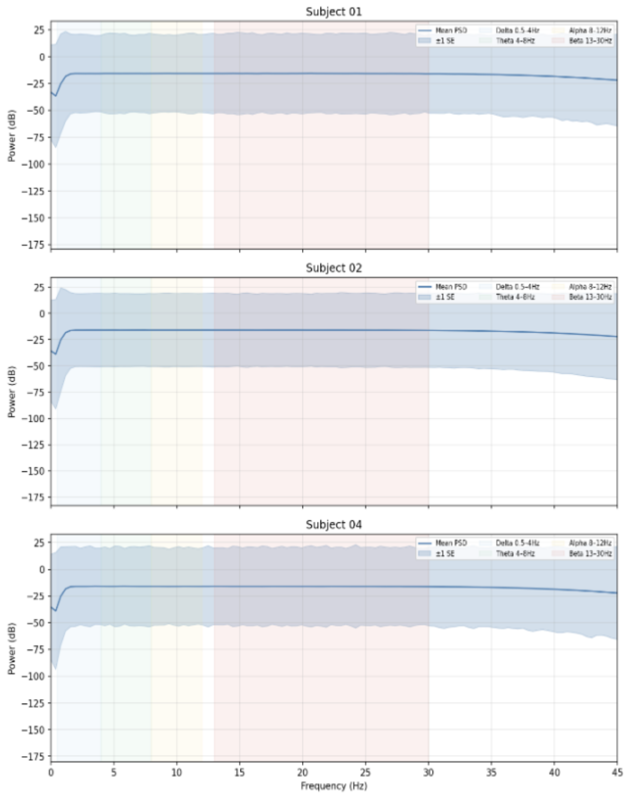

Figure 2- EEG Power Spectral Density (PSD)

The power spectral density (PSD) measures how the energy of a signal is spread out over different frequency bands. We use PSD analysis here to make sure that the pre-processed EEG signals have frequency characteristics that make sense from a physiological point of view and that the preprocessing pipeline has not added any artefacts or discontinuities (Schomer and Silva, 2018).

Figure 2 presents the mean PSD ± standard error across all 15 motor channels for each subject, computed after bandpass filtering between 1 and 45 Hz. All three subjects generate smooth, continuous PSD profiles throughout the entire frequency spectrum, devoid of artefact spikes or sudden discontinuities, thereby validating that the filtering and referencing processes yielded clean signals. Subject 01 has a wider standard error band than Subjects 02 and 04. This means that the amplitude varies more between trials, but the mean profile stays the same shape. The lack of spectral artefacts and the uniformity of the profile among subjects validate the proper functioning of the EEG preprocessing pipeline, yielding signals appropriate for deep learning analysis.

4.1.3 Raw EMG + Envelope Example

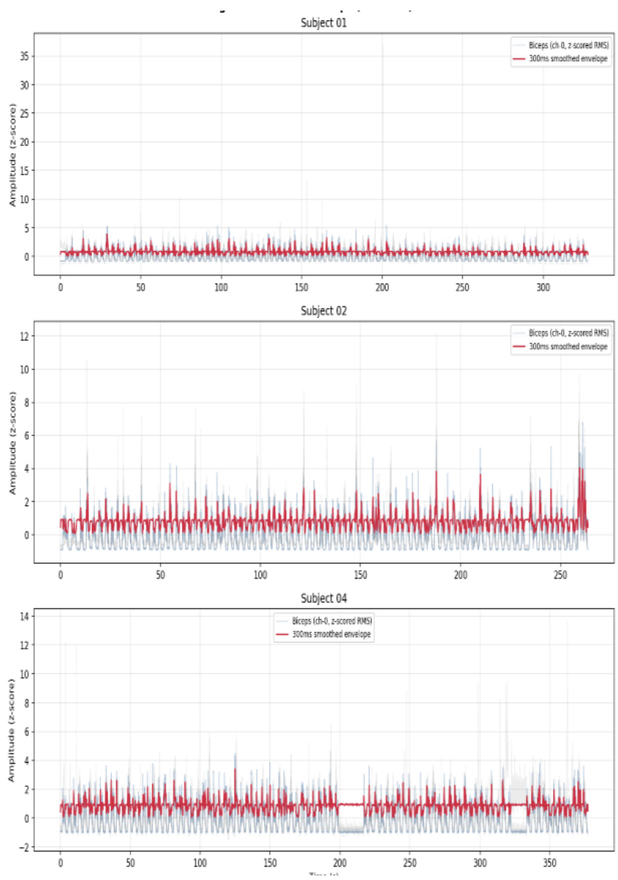

Figure 3 - Raw EMG + Envelope Example for Subjects 1-3

The EMG RMS envelope visualises muscle activation patterns across the full recording session. It is used to confirm that the EMG recordings contain clear, repeating bursts of muscle contraction corresponding to reaching trial execution, with low background noise between trials.

Figure 3 shows the z-scored RMS envelope for the biceps channel across the full session for each subject. Consistent, rhythmic bursts of activation are visible throughout all three recordings, corresponding to the repeated reaching movements. The 300ms smoothed envelope (red) makes it easy to see when each contraction period starts and ends against a stable low-amplitude baseline. Subjects 01 and 02 exhibit remarkably consistent burst patterns during the session. Subject 04 shows a short flat area between 200 and 260 seconds, which is consistent with a condition transition or rest block in the recording protocol. After that, normal activation patterns start up again. The EMG data shows strong, clear muscle activations with little noise overall, which shows that it is good for envelope-based feature extraction and classification.

4.1.4 EEG vs EMG Frequency Distribution Comparison

The frequency distribution comparison aligns both modalities on a singular spectral axis to validate their capture of physiologically distinct processes across varying frequency ranges, thereby establishing the essential rationale for their integration within a fusion framework.

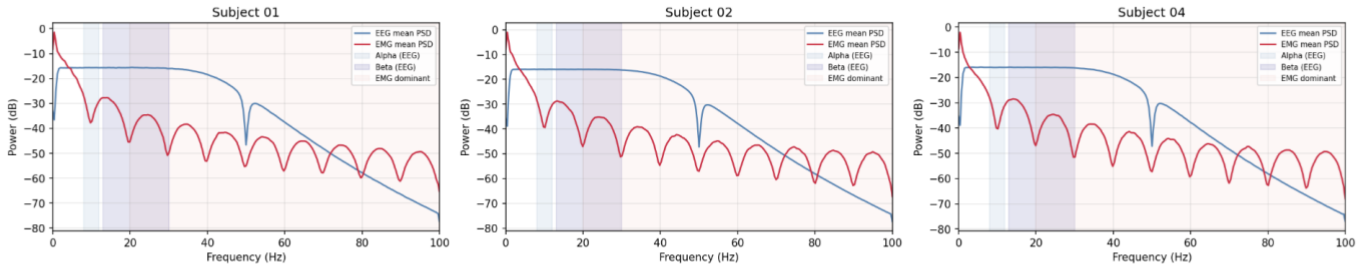

Figure 4- EEG vs EMG Frequency Distribution Comparison

Figure 4 displays the average Power Spectral Density (PSD) for EEG (blue) and EMG (red) within the 0–100 Hz frequency range for all three subjects. The outcomes are uniform among subjects. EEG power is mostly in the low-frequency range, and it drops off smoothly after 40 Hz. The alpha and beta bands are clearly in the area with the most EEG energy. In contrast, EMG power is spread out over a much wider range of high frequencies. It has periodic notches from the notch filter and a lot of activity that goes well beyond 40 Hz into the range where EEG power has mostly faded.

There is a clear spectral separation in all three subjects. The EEG power is mostly in the low-frequency range and goes down well before the frequency range where EMG activity is still strong. This confirms that EEG and EMG record different physiological processes: cortical neural preparation and peripheral muscular execution, respectively. This is the theoretical basis for multimodal fusion.

4.1.5 Conclusion of the analysis

The four preliminary analyses together show that the proposed pipeline is technically feasible. The correlations between EEG channels show that motor channels are organised in a structured way that is the same for all subjects. The PSD analysis shows that the signals are clean, free of artefacts, and have the expected frequency characteristics. The EMG envelope shows that the recordings of muscle activation are reliable and of high quality, with a clear trial structure. The frequency comparison shows that the two modalities are spectrally complementary. These findings collectively confirm the NeBULA dataset's appropriateness for the deep learning fusion framework.

4.2 Technical Feasibility

The technical feasibility of this project is supported by a robust and accessible software setup. All implementation and analysis were performed in Python 3. NumPy and Pandas were used to manipulate data, PyTorch was used to build deep learning models, MNE-Python was used to preprocess EEG data by filtering and choosing channels, and SciPy was used for signal and spectral processing. All code and version control are maintained on GitHub.

All the documentation and draft work is maintained in the OneDrive project workspace: muskaan_garg_dissertation.

4.3 Ethical and Legal Considerations Regarding Dataset

This project ensures full compliance with ethical and legal standards by using the NeBULA dataset (Garro et al., 2025), an open-access EEG-EMG recording published in Nature Scientific Data. All recordings are fully anonymised, containing no identifiable participant information, and were collected under institutional ethical approval by the original authors. The dataset is openly licensed, permitting reuse for non-commercial academic research with appropriate citation.

All analyses conform to principles of data integrity, including full transparency of methods and analyses and ensuring reproducibility. The study does not involve medical diagnosis, patient data, or patient experiments; findings are intended solely for algorithmic evaluation and methodological research within signal processing and machine learning.

4.4 Project Significance and Expected Outcomes

This project is distinctive for its reproducible evaluation of EEG–EMG fusion strategy applied for upper-limb gesture classification on an open access, synchronized dataset. As opposed to earlier work which relied on isolated or proprietary data, this project follows a transparent methodology combining spatial–temporal models (EEGNet and EMGNet) and sequential modelling (CNN-LSTM), enabling fair benchmarks of unimodal and multimodal performance (Lawhern et al., 2018; Sakhavi, Guan and Yan, 2018). Additionally, this project explores the early signal detection of EEG signals and the late detection in EMG signals, contributing to the understanding of the neural–muscular timing sequence.

The expected outcomes of this project are to establish a rigorous, reproducible benchmark comparing single-modality EEG, single-modality EMG, and multimodal fusion models under a subject-independent evaluation protocol. The results are expected to assist in understanding how each type of signal affects movement classification, how cortical and muscular signals work together over time when reaching with the upper limbs, and what conditions make multimodal fusion better than single-modality methods. The project collectively establishes a clear and reproducible framework for subsequent EEG-EMG fusion research utilising a standardised dataset.

5. Implementation and Contributions

This chapter presents the complete technical implementation of the EEG-EMG fusion pipeline. **Section 5.1** provides an overview of the full system architecture and module structure. **Section 5.2** describes the data preprocessing and window construction pipeline. **Section 5.3** details the subject-independent split strategy and its methodological justification**. Section 5.4** presents the diagnostic analysis pipeline for both EMG and EEG, with Section 5.4.1 covering EMG diagnostics and Section 5.4.2 covering the ten-method EEG diagnostic analysis. **Section 5.5** describes the EMG model development process, covering EMGNet and EMG CNN-LSTM. **Section 5.6** describes the EEG model development, covering EEGNet and EEG CNN-LSTM. **Section 5.7** presents the fusion model architecture and design rationale. Section 5.8 describes the timing analysis pipeline. **Section 5.9** summarises the chapter's key contributions.

5.1 Overall System Overview

The project is comprised of a fully functional EEG-EMG fusion pipeline that includes the entire process from raw BIDS-formatted NeBULA recordings through to the trained models, diagnostic tests, and timing results. The pipeline is composed of five distinct modules, each corresponding to a folder in the project repository.

The first module is the **preprocessing and windowing pipeline** (epoch.py), which loads raw EEG .vhdr and EMG .csv files for all subjects, applies pre-processing, and produces the windowed arrays which are used by all downstream modules. These are saved in preprocessed folder.

The second module is the **EDA and diagnostics pipeline** (eda/), which works on the windowed arrays to characterise signal quality and class separability. This module runs independently of model training and aids in model creation and for better interpretation of results.

The third module is the **subject-independent model training pipeline** (models_sub_independant/), which contains the five model scripts and their results.

The fourth module is the **subject-dependent model training pipeline** (subject_dependant/), which trains EEGNet and EMGNet per subject and their results. This module exists specifically to cross-check the EEG results from the main pipeline.

The fifth module is the **timing analysis pipeline** (temporal_analysis/), which operates on per-subject trial-level arrays and quantifies the EEG-to-EMG temporal delay across subjects and tasks.

The diagram for the overall system overview is given in *Appendix A*.

5.2 Data Pipeline and Window Construction

The preprocessing and windowing pipeline is implemented in epoch.py. It processes the free-condition recordings and produces the arrays used by all five models and the timing analysis.

EEG preprocessing: For each subject, the 128-channel BrainVision .vhdr file is loaded using MNE-Python. Fifteen motor-relevant channels covering the sensorimotor cortex bilaterally are selected: C3, C4, Cz, FC3, FC4, CP3, CP4, C1, C2, C5, C6, FC1, FC2, CP1, CP2. A 4th-order Butterworth bandpass filter (1–45 Hz) is applied via MNE's IIR filter, followed by a 50 Hz notch filter. Common average referencing (CAR) is applied to suppress volume-conduction artefacts. Signals are resampled to 200 Hz using MNE's built-in resampler, then linearly detrended and z-score normalised per channel across the full recording.

EMG preprocessing: The 11-channel EMG CSV is loaded and any flat channels are discarded. A 4th-order Butterworth bandpass filter (20–400 Hz) is applied using scipy.signal.sosfiltfilt, followed by a 50 Hz notch filter. Signals are full-wave rectified and smoothed using a 100ms RMS envelope window via scipy.ndimage.uniform_filter1d. Resampling to 200 Hz is performed by taking every fifth sample from the 1000 Hz signal. The 100ms RMS smoothing applied before decimation acts as an effective low-pass filter, substantially attenuating high-frequency content prior to downsampling. Z-score normalisation is applied per channel.

Epoch extraction: For each G event marker in the events TSV, a 2500ms epoch is extracted spanning −500ms to +2000ms around the movement onset, producing 500 samples at 200 Hz (onset at sample 100). Per-subject trial arrays of shape (n_trials, 15, 500) for EEG and (n_trials, 11, 500) for EMG are saved to preprocessed/sub-XX/ for use by the timing analysis pipeline.

Windowing: A sliding window of 80 samples (400ms) with a step of 40 samples (200ms) is applied to each epoch, yielding 11 windows per trial at 50% overlap. Fifty percent overlap is standard practice in EEG and EMG classification as it captures patterns that straddle window boundaries and doubles the number of training samples per trial (Schirrmeister et al., 2017). Each window inherits the label of its parent trial. The combined windowed arrays saved are:

- X_eeg_win.npy - (11,803, 15, 80)
- X_emg_win.npy - (11,803, 11, 80)
- y_win.npy, subject_ids_win.npy, trial_ids_win.npy - (11,803,)

The subject ID and trial ID arrays are essential for the subject-level split and for trial-aware sequence construction in the EMG models. The pre-processed windowed arrays are saved in preprocessed folder.

5.3 Subject-Independent Split Strategy

All five models use an identical subject-level data split, ensuring that results are directly comparable. The 36 subjects are randomly partitioned into training (25 subjects), validation (5 subjects), and test (6 subjects) sets, approximating a 70/15/15 split. The split is produced by calling np.random.seed(42) before shuffling the subject array, making it fully deterministic and identical across all five training scripts. The resulting test subjects are: 8, 9, 17, 24, 30, and 33.

Subject-independent evaluation is used because it reflects the real-world scenario in which a trained model must generalise to a new, unseen user without retraining. This follows the primary requirement for practical BCI deployment (Lotte et al., 2018). The use of subject-independent evaluation eliminates the possibility of data leakage into either the training or validation sets from a given test subject. As such, results will reflect true cross-subject generalization.

To provide additional context, subject dependent models were also created for both EEGNet and EMGNet. In the subject-dependent setting, each subject's data is split chronologically: the first 70% of their windows are used for training, the next 15% for validation, and the final 15% for testing. This uses temporal ordering rather than random shuffling to respect the natural progression of a recording session.

5.4 Diagnostic Analysis Pipeline

5.4.1 EMG Diagnostics

The EMG diagnostic pipeline (eda/emg_diagnostics.py) was built to understand the EMG data better to increase the accuracy of the model classification. The most relevant results are shown in *Figure 5.*

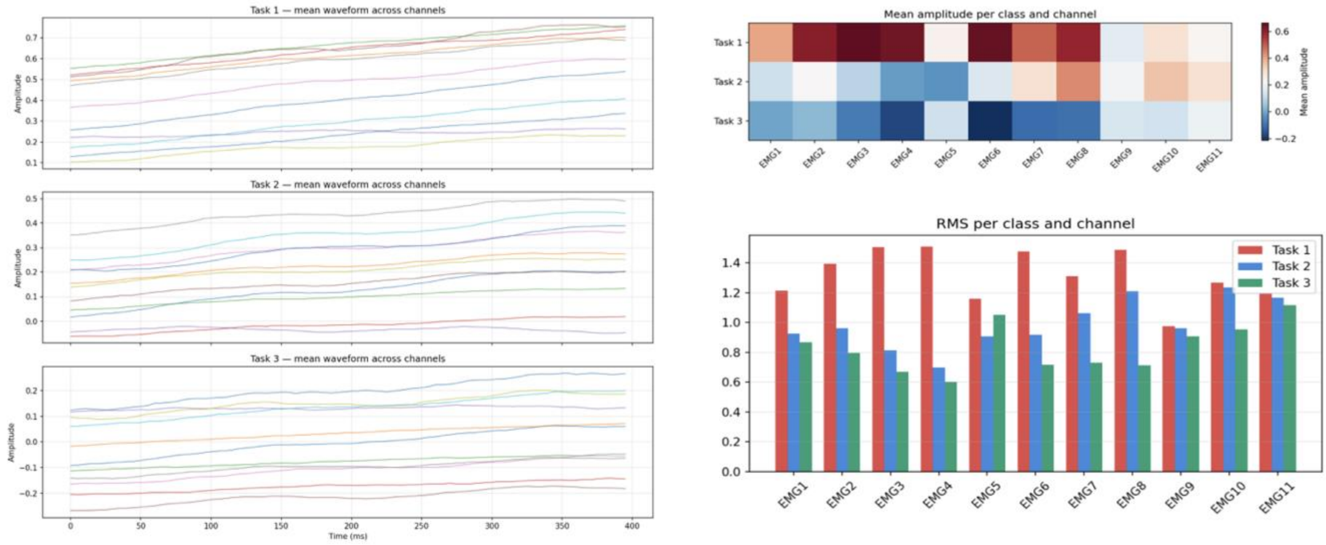

Figure 5- EMG Diagnosis Results
(a) Mean Waveform (Task 1–3)
(b) Mean Amplitude Heatmap
(c) RMS Bar Plot

The most important finding came from the mean EMG activation waveform per task across all 11 channels (*Figure 5a*). A clear and consistent amplitude ordering is visible across the execution phase of the movement. This ordering is consistent throughout the execution window and across channels, demonstrating that the three activities generate unique and consistent patterns of muscle activation throughout movement.

In the mean amplitude heatmap (*Figure 5b*) across tasks and channels, we can see that Task 1 is strongly positive (dark red) across most channels, while Task 3 is negative (dark blue), with Task 2 falling between them. Confirming that class-discriminative information is dispersed throughout all 11 EMG channels rather than focused in a subset, the contrast is significant and systematic.

Figure 5c shows the RMS amplitude and signal power per class per channel. Task 1 consistently produces the highest RMS and power across all 11 channels, with Tasks 2 and 3 showing progressively lower values.

These findings were used to make two specific changes to the design of the EMG models. Since the amplitude differences are concentrated during the active movement period, pre-movement windows were eliminated and execution-phase windows were kept. Second, instead of requiring the CNN to rediscover these amplitude-based class differences from raw signals, the model was given a direct shortcut to them by adding a manually constructed RMS feature branch to the EMG CNN-LSTM.

5.4.2 EEG Diagnostics

To understand how to build the EEG models to avoid getting low accuracy results, a diagnostic pipeline was implemented in eda/eeg_diagnostics.py. All analyses were performed independently of model training.

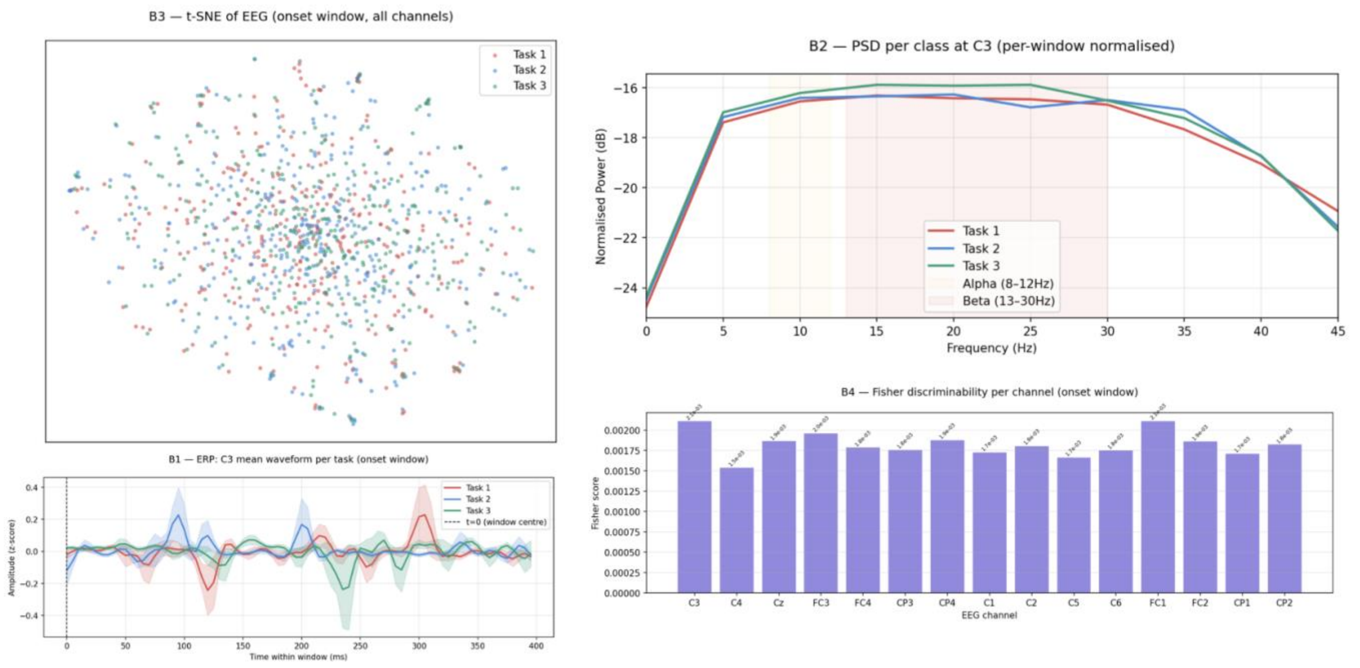

Figure 6- EEG Diagnostics Results
(a) t-SNE projection of EEG
(b) Power spectral density (PSD) per class at C3
(c) ERP mean waveform across tasks at C3
(d) Fisher discriminability score per EEG channel

Signal quality: *Figure 6c* shows the event-related potential (ERP) at channel C3 averaged across all trials per task. The three waveforms show no consistent differences - small deviations at around 100ms and 300ms are noisy and do not repeat reliably, with wide variance bands overlapping throughout. There is no stable class-discriminative component at any timepoint. *Figure 6b* also shows the power spectral density per class after per-window normalisation. The three spectral profiles overlap closely across the full 1–45 Hz range, including within the alpha and beta bands where motor-related cortical activity would be expected if the tasks produced distinguishable neural patterns (Pfurtscheller and Lopes da Silva, 1999). The absence of any spectral separation confirms no meaningful frequency-domain differences exist between the three reaching variants.

Class separability: *Figure 6a* shows a t-SNE projection of onset-window EEG features from all subjects. The three classes are uniformly mixed with no visible cluster structure - if EEG features carried any task-specific information, even partial grouping would be expected. In *Figure 6d*, we can also see that the multiclass Fisher discriminability scores for all 15 motor channels are ranging from approximately 1.5×10^{-3} to 2.1×10^{-3}. These uniformly low values indicate the three task classes are statistically indistinguishable in the EEG feature space across all channels.

Subject-dependent cross-check: To determine whether the poor results reflected a cross-subject generalisation problem or a more fundamental signal issue, EEGNet and EMGNet were trained in a subject-dependent setting using each subject's own data. EEGNet achieved 35.1% ± 16%, still at chance - while EMGNet achieved 64.8% ± 14%. EEGNet remaining near chance even within subjects rules out cross-subject variability as the explanation. This shows that the failure is a signal problem, not a generalisation problem. Full subject-dependent results are provided in *Appendix C.1*.

Window, phase, and all-channel analyses: A cross-subject logistic regression baseline evaluated each window position independently. The best single window position performed at macro-F1 = .397, while the best three-position sequence performed at macro-F1 = .361; both of these values are very close to chance. This confirms that the discriminative failure is consistent across the entire epoch. Fisher scores computed across all 128 raw EEG channels confirmed no region of the scalp gave high discriminability, with a maximum of 2.55×10^{-6}. This rules out the possibility that the 15-channel selection missed useful information.

All analysis point to the same conclusion. Combined with the NeBULA authors' own observation that no robust ERD/ERS trends were observed in the dataset (Garro et al., 2025), there is strong evidence that EEG task-discriminability is the fundamental limiting factor. The poor performance of the EEG models shows the nature of the signal, not architectural or preprocessing decisions. This finding directly informed the asymmetric gated fusion design described in Section 5.7 where EEG is treated as a potentially uninformative auxiliary signal rather than an equal contributor

5.5 EMG Model Development

Two EMG models were developed: EMGNet, a three-block 1D CNN baseline, and EMG CNN-LSTM, a hybrid model combining a raw signal branch with a handcrafted feature branch. Both were designed around the findings from the diagnostic analysis which is that the task-discriminative amplitude differences are concentrated mostly in the execution phase of the movement.

The architecture of both the models is given in Figure 7.

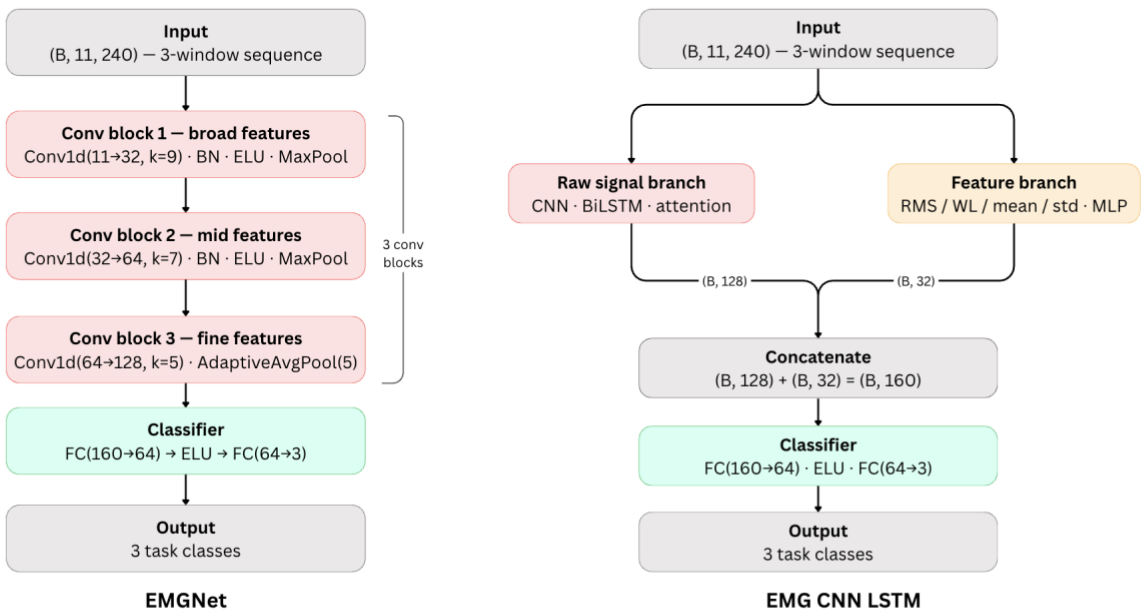

Figure 7- Architecture of EMG Models

Window selection: Initial experiments using all 11 windows per trial showed lower validation performance than expected. Diagnosis, as discussed in *Section 5.4.1,* revealed that pre-movement windows contain minimal amplitude differences across the three tasks, while execution-phase windows had consistent task-specific activation patterns. Thus, six execution-phase windows were retained for all EMG models, removing the five pre-movement and late recovery windows.

Sequence construction: Rather than treating each 80-sample window independently, three consecutive execution-phase windows are concatenated along the time axis to form a 240-sample sequence: $(11, 80) \times 3 \rightarrow (11, 240)$. Sequences are constructed trial-by-trial, grouping by (subject, trial) before concatenation to prevent sequences from crossing trial boundaries. Windows within each trial are sorted chronologically. Sliding over six execution windows with a length-3 window produces four sequences per trial. This trial-aware construction preserves within-trial temporal ordering and ensures all three windows in a sequence belong to the same movement.

5.5.1 EMGNet

EMGNet was designed as a clean baseline to see how well a straightforward convolutional approach could classify the three reaching tasks using only the raw EMG signal.

The model takes a 240-sample concatenated sequence of three execution-phase windows as input, shaped as (B, 11, 240). Three convolutional blocks are applied in sequence, each using progressively smaller kernels - 9, 7, and 5 samples respectively - to capture muscle activation features at decreasing timescales, from broad activation bursts down to finer temporal patterns. Each block includes batch normalisation and ELU activation to stabilise training, with max pooling after each of the first two blocks to reduce the time dimension. The third block uses adaptive average pooling to fix the output to a constant length of five timesteps regardless of input size, which was necessary for compatibility with Apple MPS hardware. The flattened output is passed through a two-layer fully connected classifier. Inverse-frequency class weights were applied in the loss function to account for minor class imbalance across the three tasks. Training used AdamW with ReduceLROnPlateau scheduling and early stopping on validation macro F1.

5.5.2 EMG CNN-LSTM

EMGNet treated each position in the sequence independently through convolution. The EMG CNN-LSTM was built on the observation that muscle activation during reaching is not static - it builds and evolves across the movement. A model that can track how activation patterns change over time, rather than just detecting local features, would better capture this temporal structure. Two additions were made to address this.

The first is a bidirectional LSTM with attention pooling added after the CNN backbone. The raw signal branch applies two Conv1d blocks - the same kernels as EMGNet - to extract local temporal features, producing a sequence of 60 feature vectors. This sequence is passed to a bidirectional LSTM, which processes the sequence in both forward and backward directions to capture dependencies across the full execution window. Soft attention pooling then computes a weighted sum over the 60 timesteps, allowing the model to focus on the most informative moments within the sequence, and produces a 128-dimensional embedding.

The second addition is a parallel handcrafted feature branch, motivated directly by the diagnostic finding in Section 5.4.1 that RMS amplitude differences across tasks are large, consistent, and present across all 11 channels. Rather than requiring the CNN to rediscover these patterns from scratch, a feature branch was added that computes four time-domain features per channel - RMS, mean activation, standard deviation, and waveform length (Hudgins, Parker and Scott, 1993) - across all three windows in the sequence, producing a 132-dimensional feature vector. These features are processed by a small MLP into a 32-dimensional embedding. RMS in particular is a well-established measure of muscle activation intensity and has been shown to be strongly correlated with force output during voluntary movement (De Luca, 2002), making it a reliable discriminator for tasks that differ in activation level.

The 128-dimensional raw signal embedding and the 32-dimensional feature embedding are concatenated and passed to the final classifier. Mild data augmentation - small Gaussian noise and per-channel amplitude scaling - was applied only during training to improve generalisation across subjects without distorting the amplitude relationships the model relies on.

5.6 EEG Model Development

Two EEG architectures were implemented: EEGNet (Lawhern et al., 2018) and an EEG CNN-LSTM. Both operate on single onset-centred windows treated as independent inputs, as the diagnostic analysis (*Section 5.4.2*) confirmed that 3-window sequences did not improve over single windows for EEG - a finding that itself informed the final model design.

The architecture of both the developed models is given in Figure 8.

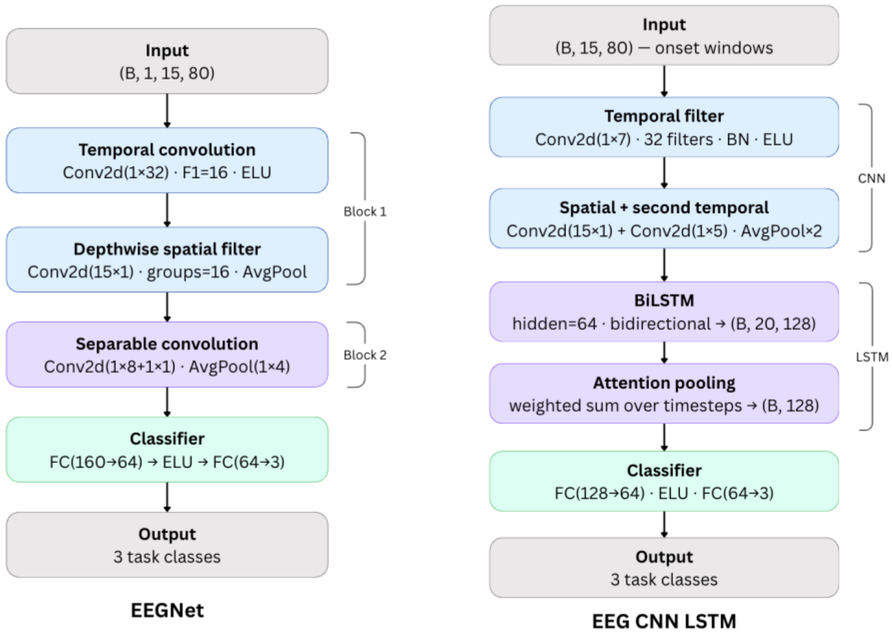

Figure 8- Architecture of EEG Models

Window selection: Three onset-centred windows were retained per trial: starting positions 40, 80, and 120 (approximately −300ms, −100ms, and +100ms relative to movement onset). These windows were identified through diagnostic analysis as having the highest, albeit still near-chance, discriminative content. Windows are used as independent single inputs to EEG models; no sequence construction is applied.

5.6.1 EEGNet

EEGNet uses a compact design which also reduces the risk of overfitting compared to larger architectures.

The model takes a single onset-centred window as input, shaped as (B, 1, 15, 80) - one channel dimension wrapping the 15 EEG channels and 80 time samples. It processes this through two blocks. Block 1 first applies a temporal convolution across time to learn frequency-specific patterns, then a depthwise spatial convolution across all 15 channels to learn which electrodes tend to activate together. Keeping these two operations separate - rather than combining them into a single layer - is the key design principle of EEGNet, as it reduces the number of parameters while preserving the ability to learn both when and where activity occurs. Average pooling then compresses the time dimension. Block 2 applies a separable convolution - a depthwise followed by a pointwise operation - which further reduces parameters without losing representational capacity, followed by a second round of pooling. The output is flattened and passed through a two-layer fully connected classifier. Hyperparameters follow the recommendations in Lawhern et al. (2018) for small EEG datasets: F1=16, D=2, F2=32. Training used AdamW with label smoothing and ReduceLROnPlateau scheduling, with early stopping monitoring validation macro F1.

5.6.2 EEG CNN-LSTM

The EEG CNN-LSTM was used to see if adding a recurrent component will improve EEGNet by explicitly modeling how patterns evolve over the 80-sample frame over time. The argument was that even if class differences in EEG are minor, a model that tracks temporal dynamics may uncover patterns that a purely convolutional model does not (Sakhavi, Guan, and Yan 2018).

The architecture begins with a CNN backbone that applies a temporal convolution followed by a spatial convolution collapsing all 15 channels, along with two rounds of average pooling reducing the time dimension from 80 to 20 steps. The result is passed to a bidirectional LSTM, which processes the 20 timesteps in both forward and reverse directions to produce a 128-dimensional representation. Soft attention pooling then computes a weighted sum over the 20 timesteps, allowing the model to focus on the most informative moments within the window, before passing to the two-layer classifier. Dropout was increased slightly compared to EEGNet to counteract the higher capacity of the recurrent component.

5.7 Fusion Model Development

The fusion model addresses RQ1 by combining EEG and EMG representations. A gated design was chosen, with the EMG branch producing the principal classification output and the EEG branch providing a correction signal controlled by a scalar sigmoid gate.

The diagnostic analysis in *Section 5.4.2* revealed that EEG does not carry a consistent task-discriminatory signal across subjects for these reaching tasks. This discovery ruled out simple feature concatenation as a design option: equal-weight fusion would require the model to rely on EEG data, risking the erosion of the robust EMG baseline. Instead, an asymmetric gated design was developed, with the EMG branch providing the major classification output and the EEG contributing only a learnt corrective signal.

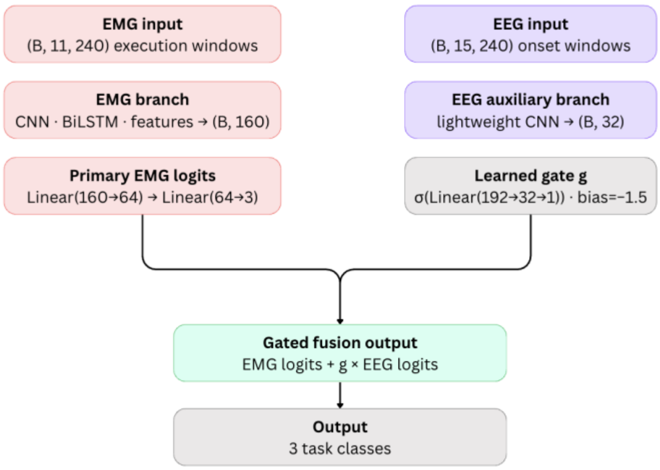

Figure 9- Architecture of the Fusion Model

The fusion model, EMGAnchoredFusion, has five components as shown in Figure 9.

The EMG raw branch mirrors the raw signal branch of the standalone EMG CNN-LSTM exactly - the same two Conv1d blocks followed by a bidirectional LSTM and attention pooling - producing a 128-dimensional embedding from the (B, 11, 240) execution-phase sequence. Reusing this design was a deliberate choice: the EMG CNN-LSTM had already demonstrated strong performance on this task, so there was no reason to redesign the EMG representation for fusion.

The EMG feature branch processes the same four handcrafted features used in the standalone model - RMS, mean, standard deviation, and waveform length - but computed for a single sequence rather than across three windows, producing a 44-dimensional vector. This is passed through a small MLP into a 32-dimensional embedding. The two EMG embeddings are concatenated into a 160-dimensional joint representation, which is passed to a two-layer classifier to produce the primary EMG logits. These logits represent the model's best estimate of the task class based on EMG alone, before any EEG contribution is applied.

The EEG auxiliary branch is kept deliberately lightweight. It processes the three concatenated onset windows - forming a (B, 15, 240) input - through a small EEGNet-style CNN with a temporal convolution followed by a spatial convolution collapsing all 15 channels, and two rounds of average pooling. The output is projected to a 32-dimensional embedding via a linear layer, and a small two-layer correction head produces EEG logits. The branch is intentionally smaller than the standalone EEG models to prevent EEG from dominating the fusion output even if the gate opens - the diagnostic analysis in *Section 5.4.2* gave no reason to trust EEG as a strong signal.

The gate is a two-layer network that takes the full 192-dimensional concatenation of the EMG representation (160-dim) and EEG embedding (32-dim) as input and produces a single scalar in [0, 1] via a sigmoid activation. The gate bias was initialised to −1.5, giving an initial output of approximately 0.18, so EEG begins mostly muted at the start of training. The model must learn during training whether opening the gate improves performance. The final output is:

output = EMG logits + g × EEG logits

If the gate remains near zero throughout training, the output is effectively EMG logits alone. This graceful degradation property was the primary reason for choosing this design over concatenation.

EMG uses the same execution-phase windows as the standalone EMG models. EEG uses the same onset-centred windows as the standalone EEG models. The same Gaussian noise and amplitude jitter augmentation from the EMG CNN-LSTM was applied to the EMG branch during training. Training used AdamW with square-root inverse-frequency class weights, label smoothing, ReduceLROnPlateau scheduling, and early stopping on validation macro F1.

5.8 Timing Analysis Pipeline

The timing analysis pipeline, which is included in temporal_analysis/temporal_pipeline.py, addresses RQ3 by calculating the trial-level temporal delay between cerebral motor preparation (EEG) and muscle activation (EMG). The study included 36 participants from all trials of the free condition.

The pipeline is subject-specific, with EEG ERD and EMG onset recognized first for each trial. These trial-level estimates are then averaged by subject, task, and condition to generate cohort-level summary. This technique preserves individual temporal variability while allowing for robust group-level interpretation.

5.8.1 EEG ERD onset detection

For each trial, beta-band power (13–30 Hz) was extracted using a fourth-order Butterworth bandpass filter, followed by the Hilbert transform to obtain the instantaneous power envelope. The signal was averaged across seven central motor channels (C3, C4, Cz, FC3, FC4, CP3, CP4) to obtain a representative cortical activity profile.

A baseline was defined over the pre-movement interval (−500ms to −100ms). ERD onset was identified as the first time point at which the beta power decreased by at least 10% relative to baseline and remained below this threshold for a minimum of 30ms.

Beta ERD - characterised by a reduction in sensorimotor beta power preceding voluntary movement - is a well-established neurophysiological marker of motor preparation (Pfurtscheller and Lopes da Silva, 1999, Kilavik et al., 2013).

5.8.2 EMG onset detection

The EMG signal was calculated as the mean RMS envelope for all 11 recorded muscles and smoothed using a 20ms moving average. Using the same baseline window as EEG, onset was defined as the first time the signal surpassed the baseline mean by more than two standard deviations and persisted above that threshold for at least 25 milliseconds.

As with EEG, the prolonged threshold requirement assures that observed onsets are actual muscle activation rather than fleeting artifacts or noise.

5.8.3 Delay and cross-correlation

For each trial, the EEG-to-EMG delay was calculated as the difference between EMG and EEG ERD onsets. A positive delay suggests that muscle activation occurs after cortical preparation, in accordance with expected neurophysiological ordering.

To validate the threshold-based estimates, we computed cross-correlation between the inverted EEG ERD signal and the EMG envelope over a ±600 ms window. The peak lag of the cross-correlation gives an independent estimate of temporal alignment, acting as a robustness check.

5.9 Implementation Summary and Contributions

This chapter described the complete technical implementation of the pipeline.
The key contributions are:

- A fully reproducible subject-independent preprocessing and evaluation pipeline on NeBULA covering 36 subjects and five models with a fixed deterministic split.
- Data-driven EMG model development from a basic CNN to a hybrid CNN-LSTM with handcrafted features, informed by execution-window analysis and trial-aware sequence construction.
- Systematic EEG evaluation across two architectures with ten independent diagnostic analyses confirming that task-discriminability, not architecture, was the limiting factor.
- A gated fusion architecture designed to handle asymmetric modality quality without assuming equal contribution.
- A trial-level EEG-to-EMG timing pipeline quantifying cortical-to-muscular delays across 36 subjects.

Chapter 6 evaluates all five models against the three research questions.

6. Evaluation

This chapter evaluates all five implemented models against the three research questions and provides critical interpretation of the results. Section 6.1 re-establishes the evaluation setup and test protocol. Section 6.2 presents the EMG model results and explains the performance improvement from EMGNet to EMG CNN-LSTM. Section 6.3 presents the EEG results and explains why both architectures performed at chance. Section 6.4 presents the fusion results and addresses RQ1. Section 6.5 presents the timing analysis results and their scientific implications, with Section 6.5.1 reporting the quantitative delay measurements and Section 6.5.2 interpreting the temporal relationship between EEG and EMG. Section 6.6 explicitly answers each of the three research questions. Section 6.7 concludes the chapter with a summary of the key findings.

6.1 Evaluation Setup and Overview

All five models were evaluated on the same fixed held-out test set, comprising six subjects - 8, 9, 17, 24, 30, and 33 - who were unseen during training and validation. This subject independent evaluation protocol was followed to ensure that reported evaluations represent true cross-subject generalization rather than within-subject memorisation.

The primary evaluation metrics are classification accuracy and macro-averaged F1-score. Macro F1 is weighted equally across all three classes regardless of class size, making it more informative than accuracy alone when class distributions differ slightly across subjects. Secondary measures such as per class precision, recall, and confusion matrices are used to discover systemic misclassification trends. All results are based on window-level predictions, consistent with how each model processes inputs during inference.

The task consists of three classes (Task 1, Task 2, and Task 3), with a 33.3% random chance baseline. All five models were trained using the same random seed (42) and subject split, allowing for direct comparison between architectures and modalities.

MODEL	MODALITY	ACCURACY	MACRO F1
EEGNet	EEG	33.9 %	0.305
EEG CNN-LSTM	EEG	35.2 %	0.337
EMGNet	EMG	79.9 %	0.801
EMG CNN-LSTM	EMG	84.1 %	0.842
Fusion	EEG + EMG	78.2 %	0.784

Table 3- Overall Models Results

Table 3 presents a summary of all five models evaluated on the same held-out test subjects under the subject-independent protocol. The findings show a distinct difference between the two modalities: EMG models produced high cross-subject classification, while EEG models performed at or close to chance regardless of architecture. Despite integrating both modalities, the fusion model did not outperform the best EMG-only result. The following sections examine each result in detail and interpret the findings against the three research questions.

6.2 EMG Results

MODEL	MODALITY	ACCURACY	MACRO F1
EMGNet	EMG	79.9 %	0.801
EMG CNN-LSTM	EMG	84.1 %	0.842

Table 4- EMG Models Results

6.2.1 EMGNet Results

EMGNet achieved 79.9% accuracy and a macro F1 of 0.801. Looking at the confusion matrix, Tasks 2 and 3 were classified well - 213/236 and 202/240 correctly respectively - while Task 1 showed more errors, with 73 samples confused with Task 2. This Task 1 vs Task 2 confusion is consistent with the diagnostic finding that Task 1 has the highest RMS amplitude, and when amplitude differences are more subtle at the boundary between tasks, a simpler model may not separate them reliably.

6.2.1 EMG CNN-LSTM Results

EMG CNN-LSTM improved on every metric, reaching 84.1% accuracy and macro F1 of 0.842. The per-class breakdown shows strong and balanced performance: Task 1 F1 = 0.846, Task 2 F1 = 0.807, Task 3 F1 = 0.874. Notably, the confusion between Task 1 and Task 2 was substantially reduced - Task 1 correct predictions improved from 154 to 181 - and no Task 3 samples were misclassified as Task 1 at all. Task 3 achieved the highest precision (0.868) and recall (0.879) of any class.

6.2.1 Evaluation of EMG Results

Both EMG models performed strongly, confirming that muscle activation signals carry reliable and consistent task-discriminative information for these reaching tasks. Two design improvements are responsible for the performance gain from EMGNet to EMG CNN-LSTM. The model was able to trace the evolution of muscle activation over the course of the 240-sample execution window thanks to the bidirectional LSTM with attention pooling, which captured temporal dynamics that the static convolutional baseline was unable to. The amplitude ordering, Task 1 > Task 2 > Task 3, was directly accessible through the handcrafted feature branch. The diagnostic analysis in Section 5.4.1 demonstrated that this ordering was robust and constant across all 11 channels. The combination of these two branches meant the model could use both learned temporal patterns and amplitude-based features together. Mild data augmentation during training was also helpful by preventing the model from overfitting to subject-specific amplitude scales.

6.3 EEG Results

MODEL	MODALITY	ACCURACY	MACRO F1
EEGNet	EEG	33.9 %	0.305
EEG CNN-LSTM	EEG	35.2 %	0.337

Table 5- EEG Models Results

6.3.1 EEGNet Results

EEGNet achieved 33.9% accuracy and macro F1 of 0.305 - barely above chance. The confusion matrix reveals a clear class collapse: Task 3 was almost entirely misclassified, with only 15 out of 180 samples correctly predicted (recall of 8.3%), and 109 of those predicted as Task 2. The model effectively learned to predict Task 2 for the majority of inputs, which is consistent with Task 2 being the most frequent prediction target rather than reflecting any genuine discriminative learning. The best validation checkpoint was found at epoch 4, indicating the model stopped improving almost immediately after training began.

6.3.2 EEG CNN-LSTM Results

The EEG CNN-LSTM model achieved 35.2% accuracy and macro F1 of 0.337 - a marginal improvement over EEGNet, but still at chance. The per-class breakdown shows Task 1 was the most difficult class, with a recall of only 16.4% - just 29 out of 177 samples correctly classified. Tasks 2 and 3 showed slightly better recall (43.5% and 45.6% respectively) but precision was low across all three classes, ranging from 0.341 to 0.360. The confusion matrix shows predictions spread fairly uniformly across all three classes with no strong diagonal, which is characteristic of a model that has found no reliable discriminative features and is essentially guessing.

6.3.3 Evaluation of EEG Results

Both EEG models performed near the random chance baseline of 33.3%, confirming the finding from the diagnostic analysis in Section 5.4.2 that the three reaching tasks do not produce distinguishable EEG signatures. The near-identical performance of two architecturally different models is significant. If the poor results were due to an inappropriate architecture, we would expect meaningful differences between the two. The fact that both converged to nearly identical results strongly supports the idea that the limiting element is the signal itself, not the model architecture.

As an additional check, EEG CNN-LSTM was also trained using the ten highest-scoring channels identified by the Fisher discriminability analysis in Section 5.4.2. Performance was lower than the 15 motor-channel baseline, indicating that no usable signal was recovered when the channel set was reduced according to discriminability scores.

This is in line with the ten diagnostic analyses in Section 5.4.2 that revealed no discriminative information in the EEG feature space at the time-domain, frequency-domain, or spatial level, as well as the NeBULA authors' own finding that this dataset did not exhibit any strong ERD/ERS trends (Garro et al., 2025).

6.4 Fusion Results

MODEL	MODALITY	ACCURACY	MACRO F1
EMG CNN-LSTM	EMG	84.1 %	0.842
Fusion	EEG + EMG	78.2 %	0.784

Table 6- Fusion Model Results

The fusion model combined both modalities through the EMG-anchored gated design described in Section 5.7, directly addressing RQ1.

The fusion model achieved 78.2% accuracy and macro F1 of 0.784 - a meaningful underperformance compared to the best single-modality model, EMG CNN-LSTM, which achieved 84.1%. This result directly answers RQ1: EEG-EMG fusion did not improve over single-modality EMG classification under these conditions.

Looking at the per-class breakdown, the fusion model performed worse than EMG CNN-LSTM on all three tasks. Task 1 recall dropped from 0.767 to 0.695, with 54 samples confused with Task 2 compared to 41 in the EMG-only model. Task 2 recall was 0.809, slightly better than EMG CNN-LSTM's 0.877 in absolute recall but with lower precision (0.677 vs 0.747), suggesting more false positives from other tasks being predicted as Task 2. Task 3 performance was comparable between the two models.

The result is consistent with the gated design behaving as expected in the worst-case scenario: EEG carries no useful signal, so the gate learns to remain near its initialised near-zero state rather than opening. Rather than adding complementary information, the EEG branch introduced additional noise into the training process, which slightly degraded the strong EMG representation. The gated design did effectively reduce the damage from the uninformative EEG branch, as seen by the fact that the fusion model still obtained 78.2%, within 6 percentage points of the EMG-only model. A more significant performance decline would have likely resulted from a naive concatenation method, which would have required the model to rely on EEG characteristics equally.

The key takeaway is that fusion is not inherently beneficial. When one modality lacks discriminative content, combining it with a strong modality does not improve performance - it risks degrading it. The value of fusion depends entirely on whether both modalities carry complementary information, which in this case they do not.

6.5 Timing Analysis and Interpretation

The timing analysis addressed RQ3 by characterising the temporal relationship between cortical motor preparation and peripheral muscle activation during unassisted upper-limb reaching. The pipeline quantified the EEG-to-EMG delay across 36 subjects and three task variants, providing insight into how the two modalities are temporally related and what this means for multimodal system design.

6.5.1 Temporal Pipeline Results

TASK	EEG ERD ONSET	EMG ONSET	EEG-TO-EMG DELAY
TASK 1	−5.48ms	171.95ms	179.22ms
TASK 2	−23.40ms	159.05ms	181.08ms
TASK 3	−14.49ms	174.97ms	188.84ms

Table 7- Temporal Pipeline Results

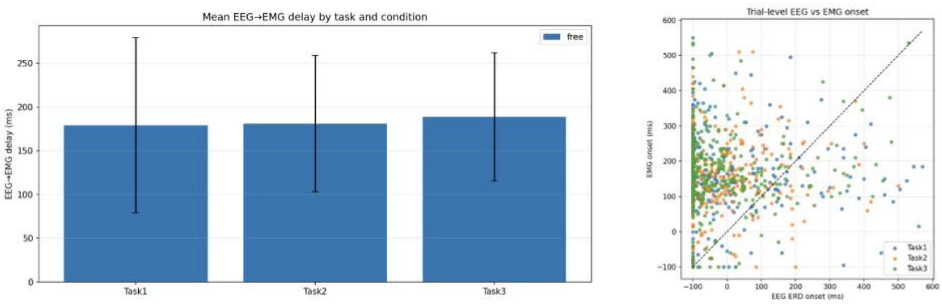

Figure 10- Temporal Pipeline Results:
(a) mean EEG–EMG delay by task,
(b) trial-level EEG vs EMG onset scatter,
(c) threshold-based vs cross-correlation delay comparison.

The results show a consistent positive delay between EEG ERD onset and EMG activation across all tasks. EEG ERD onset occurred slightly before movement (−5ms to −23ms), while EMG onset occurred substantially later (~159–175ms), indicating a clear and consistent temporal ordering (*Figure 10a*). This pattern confirms that cortical activity reliably precedes muscle activation, consistent with established models of motor control in which movement is initiated by preparatory activity in the sensorimotor cortex before descending motor commands activate peripheral muscles (Pfurtscheller and Lopes da Silva, 1999; Kilavik et al., 2013).

6.5.2 Analysis of the Temporal Relationship Between EEG and EMG

The delay is comparatively constant despite variations in movement type between tasks, indicating that the temporal gap between brain intention and execution is a basic feature of voluntary movement rather than task-specific behavior. This result is also in line with earlier studies showing that, independent of the kind of motor action, beta band desynchronization is a general preparatory mechanism involved in starting movement (Neuper, Wörtz, and Pfurtscheller, 2006; Kilavik et al., 2013). Rather than a fundamentally different brain function, the slight variation seen in Task 3 most likely reflects variations in movement complexity or execution dynamics.

The findings show that while EMG records the movement's execution phase, EEG records an initial preparatory signal. Thus, a significant physiological gap between motor planning and motor output is represented by the observed delay. The design of multimodal systems is significantly impacted by this temporal gap. In particular, it shows that whereas EMG offers trustworthy confirmation of movement execution, EEG can be employed for early identification of motor intent. According to this separation of functions, EEG and EMG should be viewed as complimentary signals that function at various stages of the motor control process rather than as redundant modalities (Farina et al., 2014).

The cross-correlation analysis (*Figure 10c*) further supports this interpretation by providing an independent estimate of temporal alignment between EEG and EMG signals. The observed temporal link is not an artifact of a particular detection technique, even though the correlation-based delays exhibit some fluctuation in comparison to threshold-based estimates. Rather, it is a reflection of a physiological link between muscle activation and brain activity. Crucially, this research shows that rather than recording specifics about the movement itself, the main utility of EEG in this situation is its capacity to record when a movement is being prepared.

This work offers a theoretical foundation for developing multimodal systems that use EMG for accurate execution tracking and EEG for anticipatory detection by explicitly quantifying the EEG-to-EMG latency. This temporal perspective offers a clear and practically useful insight into how bio signals can be integrated for improved performance in applications such as assistive technologies, rehabilitation systems, and brain–computer interfaces.

6.6 Discussion of Research Questions

RQ1 - Does EEG-EMG fusion outperform single-modality classification?

No - EEG-EMG fusion did not outperform single-modality classification. Fusion not only failed to improve over EMG-only but slightly degraded performance. This occurred because the EEG branch contributed noise rather than complementary information. The gated design limited the but the gate could not fully compensate for the absence of useful EEG signal. Fusion would have been more beneficial if both modalities carries complementary discriminative content, which was not the case for this project.

RQ2 - Which architecture performs best within each modality?

For EMG, the CNN-LSTM with handcrafted features outperformed the plain CNN baseline. The addition of temporal sequence modelling via BiLSTM and the explicit handcrafted feature branch, which provided a direct shortcut to the amplitude-based class differences identified in the diagnostic analysis, both contributed to this improvement. For EEG, both architectures performed at chance, and the marginal difference between them was not meaningful. When the signal itself contains no discriminative information, architecture choice becomes irrelevant.

RQ3 - What is the temporal delay between EEG motor onset and EMG activation?

The mean EEG-to-EMG delay was approximately 180ms across all three tasks - specifically 179ms for Task 1, 181ms for Task 2, and 189ms for Task 3. EEG beta ERD consistently preceded the movement onset marker, while EMG activation followed it, confirming the expected cortical-to-muscular temporal sequence. Inter-subject variability was high, with standard deviations in the range of 75–100ms, indicating that the absolute delay varies considerably across individuals even when the direction of the relationship is consistent.

6.7 Evaluation Summary

Across all five models evaluated on the same held-out test subjects, EMG CNN-LSTM emerged as the strongest model with 84.1% accuracy and macro F1 of 0.842, driven by temporal sequence modelling and amplitude-based handcrafted features. EMGNet also performed strongly at 79.9%, confirming that EMG alone is sufficient for reliable cross-subject task classification. Both EEG models remained at or near chance regardless of architecture, a finding supported by ten independent diagnostic analyses confirming that the three reaching tasks produce no distinguishable EEG signatures. The subject-dependent cross-check - EEGNet at 36.7% ±16.3% even within subjects - confirmed this is a signal problem, not a generalisation problem. The fusion model, despite its principled asymmetric design, did not improve over EMG-only, achieving 78.2% as EEG contributed noise rather than complementary signal. Together, these results provide a clear and consistent answer to all three research questions and motivate the broader discussion in Chapter 7.

7. Discussion and Conclusion

This chapter recontextualises the project findings, summarises contributions, reflects on limitations, and identifies directions for future work. Section 7.1 returns to the original motivation and assesses what was found relative to what was expected. Section 7.2 states the main contributions of the project. Section 7.3 discusses the key implications for researchers, model designers, and dataset curators. Section 7.4 reflects critically on the limitations of the work. Section 7.5 identifies meaningful directions for future research. Section 7.6 provides a final standalone conclusion summarising the project's primary findings and value.

7.1 Motivation and Goals Revisited

The starting point for this project was a well-established observation in the motor neuroscience and BCI literature: EEG and EMG record distinct phases of the same voluntary movement process, and their combination should, in theory, yield a more comprehensive and trustworthy picture of motor intent than either signal alone (Nicolas-Alonso and Gomez-Gil, 2012). EEG gives information of cortical preparation before movement begins, while EMG captures muscular execution after it starts, this gives a temporal complementarity that has motivated multimodal fusion research (Zhang et al., 2022; Pritchard, Campelo and Goldingay, 2025).

The expectation going into this project was that fusion would improve classification performance over single-modality baselines, and that the NeBULA dataset would provide a strong basis for evaluating this. Regardless of architecture, window selection, or assessment technique, it was discovered that EEG did not carry any task-discriminative signal for these reaching variants. As a result, fusion deteriorated rather than improved over the best EMG-only model.

This outcome is a genuine contribution rather than simply a negative result. Demonstrating rigorously - through ten independent diagnostic analyses, two EEG architectures, and a subject-dependent cross-check - that fusion fails under specific conditions is exactly the kind of systematic evidence the field needs. Assuming fusion is beneficial without verifying modality quality is a methodological gap in much of the existing literature (Sultan et al., 2025), and this project directly addresses that gap.

7.2 Main Contributions

This project made six distinct contributions beyond simply running classification experiments.

The first is a fully reproducible subject-independent deep learning benchmark on the NeBULA dataset. To the best of the author's knowledge, this is the first evaluation of deep learning models on NeBULA for upper-limb reaching classification under a cross-subject protocol. All code is publicly available on GitHub, the data split is deterministic via a fixed random seed, and all five models share identical evaluation conditions, making the results directly replicable.

The second is the systematic development of the EMG pipeline through data-driven design decisions. Based on diagnostic results, the EMG models were incrementally improved from a simple convolutional baseline. This included limiting the models to execution-phase windows, creating trial-aware sequences, and incorporating a manually created feature branch that was guided by the observed RMS amplitude ordering across tasks. This procedure resulted in a final model that achieved an impressive 84.1% macro F1 cross-subject for a three-class reaching classification issue.

The third is the ten-method EEG diagnostic analysis. This project carried out a methodical investigation covering ERP overlap, frequency-domain separability, t-SNE visualization, Fisher discriminability, per-subject analysis, window-level baselines, phase-level analysis, and all-channel Fisher scoring, as opposed to merely reporting poor EEG accuracy and attributing it to cross-subject variability or architecture limitations. This provides rigorous evidence that the failure is a fundamental property of the signal for this task.

The EEG-to-EMG timing characterization across three reaching tasks and 36 subjects in the free condition is the fourth. Together with the subject-level variability analysis, the quantification of the about 180 ms cortical-to-muscular delay provides a rigorous physiological characterization of the dataset that extends beyond classification performance.

The fifth is the evidence that fusion is not necessarily beneficial when modality quality is asymmetric. The gated fusion design explicitly tested whether a learned gate could selectively exploit EEG when useful - and the result showed it could not, because the signal itself contained nothing to exploit.

The sixth is the pipeline itself, which is open and compliant with BIDS and may be used with little modification to future EEG-EMG datasets.

7.3 Key Implications

The findings of this project carry several implications for researchers, system designers, and dataset curators.

For researchers working on multimodal bio signal fusion, the key implication is that modality discriminability must be proved before fusion is attempted. Several existing studies report fusion accuracy without first establishing that both modalities independently carry task-relevant information (Tryon and Trejos, 2021; Pritchard, 2024). This project shows that fusion is likely to deteriorate rather than improve performance when one modality is uninformative, and that the degree of degradation relies on the fusion approach employed. Before designing any fusion architecture, diagnostic analysis, like the one used here, should be a common procedure.

The findings indicate that signal quality is more important for model design than architectural complexity. Well-validated designs from the BCI literature, such as EEGNet and EEG CNN-LSTM, both performed at random, not because they were badly built but rather because there was nothing to learn from the input signal (Lawhern et al., 2018; Sakhavi, Guan, and Yan, 2018). On the other hand, data-driven design improvements significantly enhanced the EMG models. This suggests that effort spent on understanding and engineering the input representation is more valuable than effort spent on increasing model complexity.

The findings imply that EMG is a more useful signal for classification in BCI and rehabilitation system design for tasks such as these reaching variants, where cortical differentiation between movement types is limited. In paradigms with greater pre-movement cortical differentiation, such motor imagery, where tasks are intended to generate unique and repeatable neural patterns, EEG may play a more significant role (Lotte et al., 2018).

For dataset design, future EEG-EMG datasets intended for fusion research should incorporate tasks that produce distinguishable pre-movement EEG signatures. Regardless of the model employed, the dataset's usefulness for EEG-based categorization is limited since the NeBULA reaching variations are physiologically close enough that the cortical preparation signal is identical across them (Garro et al., 2025).

7.4 Limitations

The most significant limitation of this project is that EEG provided no task-discriminative signal for the three reaching variants in the NeBULA dataset. While this was rigorously confirmed through diagnostic analysis, it means that the fusion model could not be evaluated under conditions where EEG genuinely contributes - which was the original motivation for the project. It is also unclear if the gated architecture will function as planned on a dataset where EEG includes significant class-separable information, and the results characterize fusion failure rather than fusion success. This restricts the fusion conclusions' applicability to this particular dataset and task combination.

The timing analysis was conducted on the free condition only. It is yet unclear whether the EEG-to-EMG latency varies under low or high exoskeleton resistance, when movement patterns and effort levels are different. Without more research, the temporal results cannot be extrapolated to assisted reaching.

Only EEGNet and EMGNet underwent subject-dependent model validation. A more comprehensive view of how within-subject versus cross-subject performance varies between designs would be obtained by extending this to EEG CNN-LSTM, EMG CNN-LSTM, and the fusion model.

No statistical significance testing was conducted between model performance differences. Given the relatively small test set of six subjects, some of the observed differences in accuracy may not be statistically significant, and confidence intervals were not computed.

7.5 Future Work

The most immediate extension is to run the timing analysis across all three NeBULA conditions - free, low resistance, and high resistance - to quantify how exoskeleton assistance affects the EEG-to-EMG delay. If the delay increases under assisted conditions, this would have direct implications for how timing-aware fusion systems should be calibrated for exoskeleton control.

It would be possible to test whether the EMG models and fusion architecture generalize beyond reaching tasks and whether EEG can contribute significantly when the task design supports it by applying this pipeline to datasets with stronger pre-movement EEG differentiation, such as motor imagery paradigms, where tasks are designed to produce distinct cortical signatures (Lotte et al., 2018).

Subject adaptation and transfer learning approaches could improve cross-subject EMG generalisation. The subject-dependent results showed high variance (±13.2% for EMGNet), suggesting that some subjects are considerably harder to generalise to than others. Fine-tuning a cross-subject model on a small amount of target-subject data could reduce this variance without requiring full retraining.

Transformer-based or cross-attention fusion architectures could learn cross-modal temporal alignment directly from data, potentially exploiting the ~180ms EEG-to-EMG delay in a more principled way than the current gated design. This approach has shown promise in other multimodal time-series settings (Sultan et al., 2025).

7.6 Final Conclusion

This project implemented and evaluated a complete EEG-EMG deep learning pipeline on the NeBULA dataset under a rigorous subject-independent evaluation protocol. Five models were developed and assessed - two EEG architectures, two EMG architectures, and a gated fusion model - alongside a systematic diagnostic pipeline and a trial-level timing analysis across 36 subjects.

EMG CNN-LSTM achieved the strongest result at 84.1% macro F1, confirming that EMG carries reliable and consistent task-discriminative information for these reaching variants across unseen subjects. Both EEG models remained at chance regardless of architecture - a finding investigated through ten independent diagnostic analyses, all pointing to the same conclusion: the three reaching tasks do not produce distinguishable EEG signatures.

The timing analysis showed that EEG cortical preparation consistently precedes EMG activation by approximately 180ms, confirming the signals are physiologically valid and temporally complementary. However, this preparation signal is identical across all three task variants - EEG signals that a movement is coming, but not which one - which explains both the classification failure and the fusion outcome.

The fusion model did not outperform EMG-only. The learned gate suppressed EEG contribution throughout training, converging to near EMG-only behaviour. The project's primary value lies not in achieving high fusion accuracy, but in rigorously demonstrating why fusion fails under these conditions - a finding that should directly inform how future multimodal bio signal systems are designed, evaluated, and benchmarked.

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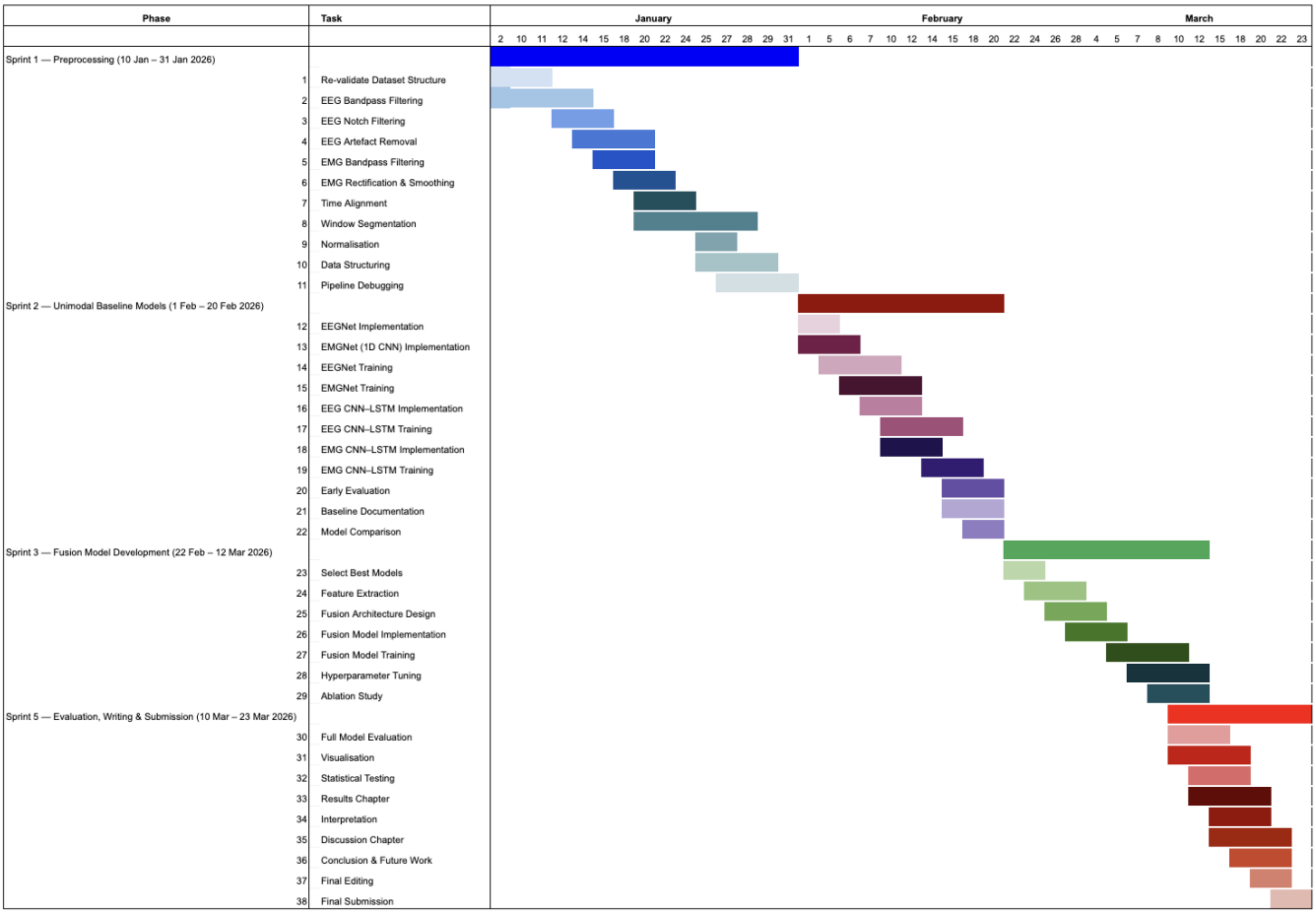
Project Management

Work plan - Gantt Chart



Risk & Mitigation table

Risk & Mitigation Table - updated to reflect what actually happened:

Risk ID	Risk Description	Impact	Likelihood	Mitigation Strategy	Outcome
R1	Incomplete or noisy EEG/EMG recordings due to sensor artefacts	High	Medium	Applied bandpass and notch filtering, common average referencing for EEG, RMS envelope smoothing for EMG. Three subjects excluded due to documented signal quality issues.	Mitigated - preprocessing pipeline produced clean signals confirmed by feasibility analysis
R2	Imbalanced gesture classes leading to biased model training	Medium	Medium	Used inverse-frequency class weights in the loss function and macro F1 as the primary evaluation metric.	Mitigated - class distribution was broadly balanced across the three tasks
R3	Overfitting in deep learning models	High	Medium	Implemented dropout, batch normalisation, early stopping on validation macro F1, and gradient clipping.	Partially mitigated - EMG models showed some train/val gap but test performance remained strong
R4	Limited computational resources for model training	Medium	Low	Training performed locally on Apple M-series hardware using the MPS backend rather than Google Colab.	Not an issue - local MPS hardware was sufficient for all five models
R5	EEG-EMG fusion model fails to outperform single modalities	Medium	High	Designed an asymmetric gated fusion architecture to limit degradation from uninformative EEG. Conducted systematic diagnostic analysis to understand EEG limitations before designing fusion.	Materialised - fusion did not outperform EMG-only, confirmed to be a signal discriminability issue rather than a fusion design failure
R6	EEG signal contains insufficient task-discriminative information	High	Low (unanticipated)	Conducted ten independent diagnostic analyses to characterise EEG discriminability and rule out architectural or preprocessing explanations.	Materialised - confirmed as the primary limiting factor for both EEG models and the fusion model

Ethical statement

This project ensures full compliance with ethical and legal standards by using the NeBULA dataset (Garro et al., 2025), an open-access EEG-EMG recording published in Nature Scientific Data. All recordings are fully anonymised, containing no identifiable participant information, and were collected under institutional ethical approval by the original authors. The dataset is openly licensed, permitting reuse for non-commercial academic research with appropriate citation.

All data handling, preprocessing, and analysis is conducted in accordance with ethical research principles outlined by Heriot-Watt University. The dataset is used solely for academic and research purposes within the scope of this dissertation, under supervised guidance.

Dataset available in: https://doi.org/10.6084/m9.figshare.27301629

GenAI Summary

I used generative AI tools strictly as permitted by the guidelines for use at Heriot-Watt University to support my work. Generative AI was used to:

- Enhance my work for readability, clarity, and grammar: I used AI techniques to improve the language quality of particular passages. I had initially written in order to make certain portions easier to read, to improve the writing's academic tone, to refine its style, and to proofread it. AI merely improved the content and style of my writing; it did not produce any new content.
- Produce ideas to aid with my research: I used AI to find several academic articles relevant to the project area, uncover some common patterns in the EEG-EMG and gesture recognition literature, and suggest potential study pathways. I didn't utilize AI to write the research paper or other written materials; rather, I used it to help identify themes or areas of inquiry
- Aid in developing a format for organizing the report: I used AI to help create an outline for the report's structure by suggesting logical titles for its sections, suggesting subsections for each of those titles, and helping to create a general flow of work. However, AI did not produce any of the written content found in these components.
- Support code debugging and refinement: I used AI to assist find and fix errors in my Python implementation, make suggestions for better code organization and readability, and clarify confusing library methods. I am solely responsible for the experimental design, model architectural choices, training methods, and outcomes.

This report's final version does not contain any content produced by an AI tool, including paragraphs of code, analysis, charts, figures, or any other type of content. This report's design, writing, experiments, and findings are all entirely my own.

Project proposal and research changes

Project Proposal and Research Changes

This section summarises the key changes made between the project proposal submitted in D1 and the final implementation presented in this dissertation. Changes were made in response to practical discoveries during implementation, improvements to the experimental design, and a switch to a higher-quality dataset.

Dataset

The most significant change from D1 was the replacement of the original dataset. D1 proposed using two small open-access datasets - an 8-channel EEG and 8-channel EMG dataset from Mendeley Data (Dere, 2022) and a similar dataset from IEEE DataPort (Lee, 2023) - covering 11 subjects performing 7 upper-limb gestures. During early implementation, both datasets presented significant signal quality issues including timestamp misalignments, poor EEG signal quality, and insufficient subject counts for subject-independent evaluation.

These datasets were replaced with the NeBULA dataset (Garro et al., 2025), a high-density EEG-EMG recording published in Nature Scientific Data. NeBULA provides 128-channel EEG and 11-muscle EMG from 40 subjects performing upper-limb reaching tasks, with hardware-synchronised recordings and full BIDS-compliant organisation. This switch substantially improved signal quality, increased the subject count to 37 usable subjects, and enabled a rigorous subject-independent evaluation protocol. The task structure also changed - from 7 discrete hand gestures to 3 variants of upper-limb reaching - which ultimately revealed that the reaching variants produced indistinguishable EEG signatures, a finding that became a central contribution of the project.

Research Questions

D1 did not explicitly state formal research questions. D3 formalised three research questions: whether fusion outperforms single-modality models (RQ1), which architecture best captures movement-discriminative patterns in each modality (RQ2), and what the temporal delay is between EEG motor onset and EMG activation (RQ3). The addition of RQ3 and the associated timing analysis pipeline was not present in D1 and represents a new contribution developed during implementation.

Preprocessing Pipeline

D1 proposed a bandpass filter of 0.5–45 Hz for EEG and 20–450 Hz for EMG, with z-score normalisation and overlapping windows of 200–500 ms. The implemented pipeline in D3 refined these choices based on the NeBULA dataset characteristics: EEG used a 1–45 Hz bandpass with common average referencing, linear detrending, and resampling to 200 Hz; EMG used a 20–400 Hz bandpass with full-wave rectification and a 100 ms RMS envelope window, resampled to 200 Hz via decimation. Epochs were extracted from −500 ms to +2000 ms around movement onset, and a sliding window of 80 samples with a step of 40 samples was applied, yielding 11 windows per trial. These are more precise and justified choices than those proposed in D1.

Fusion Architecture

D1 proposed a simple feature-level concatenation fusion model - combining the best-performing EEG and EMG embeddings into a single classifier. This approach was replaced in D3 with an asymmetric gated fusion model (EMGAnchoredFusion), where EMG produces the primary classification output and EEG contributes a correction signal controlled by a learned sigmoid gate. This design change was motivated by the diagnostic finding that EEG carried no reliable task-discriminative signal, making equal-weight concatenation likely to degrade rather than improve performance.

Evaluation Metrics

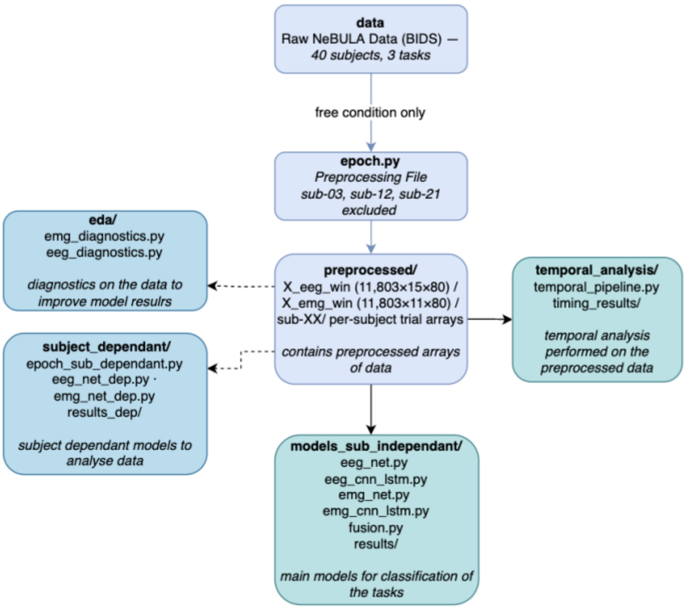
D1 proposed evaluating models using accuracy, macro F1-score, ROC-AUC, confusion matrices, and statistical significance tests (paired t-tests or Wilcoxon signed-rank tests). In D3, ROC-AUC was removed as a metric - it is more informative for binary classification problems and less standard for three-class window-level classification - and statistical significance testing between model performances was not conducted due to the small test set size of six subjects. Accuracy and macro F1 were retained as the primary metrics.

Software and Tools

D1 proposed using TensorFlow for deep learning model development and Google Colab for training. D3 used PyTorch throughout, with the AdamW optimiser rather than the Adam optimiser proposed in D1. Training was performed locally on Apple M-series hardware using the MPS backend, which proved sufficient for all five models without requiring cloud resources.

Appendix

A. Code Structure Overview



B. Subject Independent Results

B.1 EEG_Net Results

File - eeg_net

```
/Users/muskaan_garg_/GitHub/eeg_emg_fusion_ml/.venv/bin/python /Users/muskaan_garg_/
GitHub/eeg_emg_fusion_ml/models_sub_independant/eeg_net.py
========================================================================
  EEGNet — Subject-Independent NeBULA Classification
========================================================================
  Device         : mps
  Input shape    : (3219, 15, 80)
  Window starts  : [40, 80, 120]
  Subjects       : [1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14, 15, 16, 17, 18, 19, 20, 22,
23, 24, 25, 26, 27, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40]
  Train subjects : [1, 2, 4, 5, 6, 7, 10, 11, 14, 15, 16, 18, 19, 20, 23, 25, 29, 31, 34
, 35, 36, 37, 38, 39, 40]
  Val subjects   : [13, 22, 26, 27, 32]
  Test subjects  : [8, 9, 17, 24, 30, 33]
  train:  2235 windows  (Task0=735, Task1=750, Task2=750)
  val  :   450 windows  (Task0=150, Task1=150, Task2=150)
  test :   534 windows  (Task0=177, Task1=177, Task2=180)
  Parameters     : 12,931

  Epoch 001 | train loss=1.1062 acc=0.334 f1=0.314 | val loss=1.1015 acc=0.324 f1=0.268
 | lr=0.000100
  Epoch 002 | train loss=1.1020 acc=0.349 f1=0.330 | val loss=1.1021 acc=0.338 f1=0.268
 | lr=0.000100
  Epoch 003 | train loss=1.1025 acc=0.337 f1=0.320 | val loss=1.1022 acc=0.338 f1=0.269
 | lr=0.000100
  Epoch 004 | train loss=1.1045 acc=0.339 f1=0.321 | val loss=1.1025 acc=0.344 f1=0.300
 | lr=0.000100
  Epoch 005 | train loss=1.1003 acc=0.354 f1=0.341 | val loss=1.1025 acc=0.329 f1=0.260
 | lr=0.000100
  Epoch 006 | train loss=1.0995 acc=0.367 f1=0.355 | val loss=1.1024 acc=0.338 f1=0.279
 | lr=0.000100
  Epoch 007 | train loss=1.0983 acc=0.358 f1=0.347 | val loss=1.1020 acc=0.327 f1=0.266
 | lr=0.000100
  Epoch 008 | train loss=1.0983 acc=0.347 f1=0.340 | val loss=1.1022 acc=0.338 f1=0.286
 | lr=0.000100
  Epoch 009 | train loss=1.0979 acc=0.352 f1=0.341 | val loss=1.1021 acc=0.347 f1=0.291
 | lr=0.000100
  Epoch 010 | train loss=1.0964 acc=0.359 f1=0.351 | val loss=1.1020 acc=0.329 f1=0.263
 | lr=0.000100
  Epoch 011 | train loss=1.0955 acc=0.349 f1=0.335 | val loss=1.1025 acc=0.333 f1=0.263
 | lr=0.000100
  Epoch 012 | train loss=1.0943 acc=0.349 f1=0.337 | val loss=1.1032 acc=0.327 f1=0.268
 | lr=0.000100
  Epoch 013 | train loss=1.0902 acc=0.374 f1=0.366 | val loss=1.1042 acc=0.329 f1=0.286
 | lr=0.000050
  Epoch 014 | train loss=1.0917 acc=0.360 f1=0.353 | val loss=1.1034 acc=0.320 f1=0.276
 | lr=0.000050
  Epoch 015 | train loss=1.0955 acc=0.361 f1=0.352 | val loss=1.1034 acc=0.331 f1=0.272
 | lr=0.000050
  Epoch 016 | train loss=1.0908 acc=0.363 f1=0.357 | val loss=1.1031 acc=0.327 f1=0.267
 | lr=0.000050
  Epoch 017 | train loss=1.0944 acc=0.353 f1=0.346 | val loss=1.1041 acc=0.322 f1=0.271
 | lr=0.000050
  Epoch 018 | train loss=1.0932 acc=0.365 f1=0.357 | val loss=1.1037 acc=0.333 f1=0.284
 | lr=0.000050
  Epoch 019 | train loss=1.0914 acc=0.383 f1=0.378 | val loss=1.1030 acc=0.331 f1=0.278
 | lr=0.000050
  Epoch 020 | train loss=1.0912 acc=0.371 f1=0.363 | val loss=1.1035 acc=0.333 f1=0.281
 | lr=0.000050
  Epoch 021 | train loss=1.0919 acc=0.362 f1=0.355 | val loss=1.1035 acc=0.336 f1=0.282
 | lr=0.000050
  Epoch 022 | train loss=1.0930 acc=0.353 f1=0.347 | val loss=1.1041 acc=0.331 f1=0.277
 | lr=0.000025
  Epoch 023 | train loss=1.0877 acc=0.382 f1=0.373 | val loss=1.1041 acc=0.336 f1=0.287
 | lr=0.000025
  Epoch 024 | train loss=1.0882 acc=0.378 f1=0.369 | val loss=1.1044 acc=0.342 f1=0.290
 | lr=0.000025
  Epoch 025 | train loss=1.0901 acc=0.367 f1=0.359 | val loss=1.1052 acc=0.338 f1=0.289
 | lr=0.000025
  Epoch 026 | train loss=1.0914 acc=0.357 f1=0.348 | val loss=1.1052 acc=0.338 f1=0.291
 | lr=0.000025
  Epoch 027 | train loss=1.0894 acc=0.382 f1=0.375 | val loss=1.1049 acc=0.344 f1=0.294
 | lr=0.000025
  Epoch 028 | train loss=1.0887 acc=0.381 f1=0.375 | val loss=1.1052 acc=0.338 f1=0.292
 | lr=0.000025
  Epoch 029 | train loss=1.0880 acc=0.381 f1=0.375 | val loss=1.1041 acc=0.324 f1=0.276
 | lr=0.000025
  Epoch 030 | train loss=1.0896 acc=0.379 f1=0.373 | val loss=1.1044 acc=0.340 f1=0.290
```

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B.2 EMG_Net Results

File - emg_net

```
/Users/muskaan_garg_/GitHub/eeg_emg_fusion_ml/.venv/bin/python /Users/muskaan_garg_/
GitHub/eeg_emg_fusion_ml/models_sub_independant/emg_net.py
-----------------------------------------------------------------
  EMGNet — Subject-Independent NeBULA Classification
-----------------------------------------------------------------
  Device          : mps
  Input shape     : (4292, 11, 240)
  Window starts   : [120, 160, 200, 240, 280, 320]
  Train subjects  : [1, 2, 4, 5, 6, 7, 10, 11, 14, 15, 16, 18, 19, 20, 23, 25, 29, 31, 34
, 35, 36, 37, 38, 39, 40]
  Val subjects    : [13, 22, 26, 27, 32]
  Test subjects   : [8, 9, 17, 24, 30, 33]
  Train samples   : 2980
  Val samples     : 600
  Test samples    : 712
  Parameters      : 100,355

  Epoch 001 | train loss=0.5506 acc=0.757 f1=0.755 | val loss=0.3052 acc=0.893 f1=0.893
  Epoch 002 | train loss=0.4509 acc=0.813 f1=0.812 | val loss=0.3051 acc=0.892 f1=0.892
  Epoch 003 | train loss=0.3991 acc=0.830 f1=0.829 | val loss=0.2956 acc=0.908 f1=0.908
  Epoch 004 | train loss=0.3835 acc=0.840 f1=0.840 | val loss=0.3327 acc=0.878 f1=0.878
  Epoch 005 | train loss=0.3676 acc=0.841 f1=0.841 | val loss=0.2870 acc=0.898 f1=0.899
  Epoch 006 | train loss=0.3512 acc=0.850 f1=0.850 | val loss=0.2985 acc=0.903 f1=0.904
  Epoch 007 | train loss=0.3563 acc=0.848 f1=0.849 | val loss=0.2497 acc=0.912 f1=0.911
  Epoch 008 | train loss=0.3578 acc=0.851 f1=0.851 | val loss=0.2766 acc=0.902 f1=0.902
  Epoch 009 | train loss=0.3379 acc=0.852 f1=0.852 | val loss=0.2633 acc=0.910 f1=0.910
  Epoch 010 | train loss=0.3097 acc=0.873 f1=0.874 | val loss=0.2648 acc=0.915 f1=0.916
  Epoch 011 | train loss=0.3246 acc=0.862 f1=0.862 | val loss=0.2485 acc=0.925 f1=0.925
  Epoch 012 | train loss=0.3169 acc=0.870 f1=0.871 | val loss=0.3130 acc=0.885 f1=0.885
  Epoch 013 | train loss=0.3130 acc=0.873 f1=0.873 | val loss=0.2343 acc=0.928 f1=0.929
  Epoch 014 | train loss=0.2914 acc=0.882 f1=0.882 | val loss=0.2922 acc=0.905 f1=0.905
  Epoch 015 | train loss=0.3071 acc=0.873 f1=0.873 | val loss=0.3381 acc=0.880 f1=0.879
  Epoch 016 | train loss=0.2874 acc=0.885 f1=0.886 | val loss=0.2378 acc=0.918 f1=0.918
  Epoch 017 | train loss=0.3179 acc=0.865 f1=0.865 | val loss=0.3580 acc=0.857 f1=0.857
  Epoch 018 | train loss=0.2897 acc=0.876 f1=0.876 | val loss=0.2267 acc=0.923 f1=0.923
  Epoch 019 | train loss=0.2764 acc=0.886 f1=0.886 | val loss=0.2536 acc=0.918 f1=0.919
  Epoch 020 | train loss=0.2651 acc=0.891 f1=0.891 | val loss=0.2597 acc=0.913 f1=0.914
  Epoch 021 | train loss=0.2516 acc=0.891 f1=0.891 | val loss=0.2935 acc=0.913 f1=0.914
  Epoch 022 | train loss=0.2452 acc=0.900 f1=0.900 | val loss=0.2604 acc=0.922 f1=0.922
  Epoch 023 | train loss=0.2505 acc=0.894 f1=0.894 | val loss=0.2886 acc=0.893 f1=0.893
  Epoch 024 | train loss=0.2444 acc=0.897 f1=0.897 | val loss=0.2650 acc=0.923 f1=0.924
  Epoch 025 | train loss=0.2481 acc=0.896 f1=0.896 | val loss=0.2391 acc=0.933 f1=0.934
  Epoch 026 | train loss=0.2307 acc=0.902 f1=0.902 | val loss=0.2975 acc=0.912 f1=0.912
  Epoch 027 | train loss=0.2358 acc=0.904 f1=0.904 | val loss=0.2829 acc=0.913 f1=0.914
  Epoch 028 | train loss=0.2331 acc=0.907 f1=0.907 | val loss=0.3042 acc=0.905 f1=0.906
  Epoch 029 | train loss=0.2362 acc=0.908 f1=0.908 | val loss=0.3003 acc=0.913 f1=0.913
  Epoch 030 | train loss=0.2251 acc=0.907 f1=0.907 | val loss=0.2852 acc=0.907 f1=0.907
  Epoch 031 | train loss=0.2283 acc=0.902 f1=0.903 | val loss=0.3450 acc=0.907 f1=0.906
  Epoch 032 | train loss=0.2190 acc=0.915 f1=0.916 | val loss=0.2754 acc=0.913 f1=0.914
  Epoch 033 | train loss=0.2077 acc=0.911 f1=0.911 | val loss=0.3379 acc=0.907 f1=0.907
  Epoch 034 | train loss=0.2150 acc=0.911 f1=0.911 | val loss=0.2893 acc=0.913 f1=0.914
  Epoch 035 | train loss=0.2094 acc=0.912 f1=0.913 | val loss=0.2870 acc=0.920 f1=0.920
  Epoch 036 | train loss=0.2283 acc=0.910 f1=0.911 | val loss=0.2679 acc=0.920 f1=0.920
  Epoch 037 | train loss=0.2130 acc=0.912 f1=0.912 | val loss=0.3130 acc=0.913 f1=0.914
  Epoch 038 | train loss=0.1937 acc=0.924 f1=0.924 | val loss=0.2801 acc=0.922 f1=0.922
  Epoch 039 | train loss=0.2040 acc=0.915 f1=0.915 | val loss=0.3102 acc=0.915 f1=0.915
  Epoch 040 | train loss=0.2066 acc=0.911 f1=0.912 | val loss=0.2854 acc=0.922 f1=0.922

  Early stopping at epoch 40. Best epoch: 25

-----------------------------------------------------------------
  TEST RESULTS
  Best epoch     : 25
  Accuracy       : 0.7992  (79.9%)
  F1 macro       : 0.8007
  Confusion matrix:
[[154  73   9]
 [  4 213  19]
 [  0  38 202]]
-----------------------------------------------------------------

  Model   → ./results/emg_net/emg_net.pt
  History → ./results/emg_net/emg_net_history.npy
  Summary → ./results/emg_net/emg_net_summary.json

Process finished with exit code 0
```

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B.3 EEG CNN LSTM Results

File - eeg_cnn_lstm

```
/Users/muskaan_garg_/GitHub/eeg_emg_fusion_ml/.venv/bin/python /Users/muskaan_garg_/
GitHub/eeg_emg_fusion_ml/models_sub_independant/eeg_cnn_lstm.py
-----------------------------------------------------------------
  EEG CNN-LSTM — Subject-Independent NeBULA Classification
-----------------------------------------------------------------
  Device          : mps
  Input shape     : (3219, 15, 80)
  Window starts   : [40, 80, 120]
  Train subjects : [1, 2, 4, 5, 6, 7, 10, 11, 14, 15, 16, 18, 19, 20, 23, 25, 29, 31, 34
, 35, 36, 37, 38, 39, 40]
  Val subjects    : [13, 22, 26, 27, 32]
  Test subjects   : [8, 9, 17, 24, 30, 33]
  Parameters      : 101,220

  train:  2235 windows  (Task0=735, Task1=750, Task2=750)
  val  :   450 windows  (Task0=150, Task1=150, Task2=150)
  test :   534 windows  (Task0=177, Task1=177, Task2=180)

  Epoch 001 | train loss=1.1009 acc=0.346 f1=0.276 | val loss=1.1004 acc=0.336 f1=0.202
 | lr=1.00e-04
  Epoch 002 | train loss=1.0970 acc=0.346 f1=0.273 | val loss=1.0991 acc=0.327 f1=0.204
 | lr=1.00e-04
  Epoch 003 | train loss=1.0951 acc=0.338 f1=0.304 | val loss=1.0988 acc=0.313 f1=0.205
 | lr=1.00e-04
  Epoch 004 | train loss=1.0923 acc=0.341 f1=0.318 | val loss=1.0995 acc=0.309 f1=0.249
 | lr=1.00e-04
  Epoch 005 | train loss=1.0894 acc=0.361 f1=0.343 | val loss=1.1003 acc=0.327 f1=0.284
 | lr=1.00e-04
  Epoch 006 | train loss=1.0878 acc=0.367 f1=0.349 | val loss=1.1008 acc=0.322 f1=0.283
 | lr=1.00e-04
  Epoch 007 | train loss=1.0870 acc=0.364 f1=0.349 | val loss=1.1010 acc=0.322 f1=0.279
 | lr=1.00e-04
  Epoch 008 | train loss=1.0866 acc=0.355 f1=0.339 | val loss=1.1034 acc=0.338 f1=0.295
 | lr=1.00e-04
  Epoch 009 | train loss=1.0835 acc=0.373 f1=0.354 | val loss=1.1032 acc=0.311 f1=0.264
 | lr=1.00e-04
  Epoch 010 | train loss=1.0829 acc=0.389 f1=0.374 | val loss=1.1053 acc=0.344 f1=0.303
 | lr=1.00e-04
  Epoch 011 | train loss=1.0849 acc=0.369 f1=0.349 | val loss=1.1041 acc=0.313 f1=0.263
 | lr=1.00e-04
  Epoch 012 | train loss=1.0830 acc=0.387 f1=0.370 | val loss=1.1034 acc=0.313 f1=0.260
 | lr=1.00e-04
  Epoch 013 | train loss=1.0813 acc=0.374 f1=0.361 | val loss=1.1049 acc=0.316 f1=0.289
 | lr=1.00e-04
  Epoch 014 | train loss=1.0796 acc=0.374 f1=0.360 | val loss=1.1065 acc=0.304 f1=0.267
 | lr=1.00e-04
  Epoch 015 | train loss=1.0771 acc=0.377 f1=0.362 | val loss=1.1078 acc=0.307 f1=0.262
 | lr=1.00e-04
  Epoch 016 | train loss=1.0754 acc=0.389 f1=0.378 | val loss=1.1114 acc=0.316 f1=0.288
 | lr=1.00e-04
  Epoch 017 | train loss=1.0764 acc=0.389 f1=0.377 | val loss=1.1089 acc=0.353 f1=0.313
 | lr=1.00e-04
  Epoch 018 | train loss=1.0733 acc=0.386 f1=0.371 | val loss=1.1106 acc=0.309 f1=0.264
 | lr=1.00e-04
  Epoch 019 | train loss=1.0755 acc=0.368 f1=0.363 | val loss=1.1109 acc=0.320 f1=0.296
 | lr=1.00e-04
  Epoch 020 | train loss=1.0715 acc=0.399 f1=0.389 | val loss=1.1108 acc=0.318 f1=0.271
 | lr=1.00e-04
  Epoch 021 | train loss=1.0688 acc=0.407 f1=0.397 | val loss=1.1122 acc=0.316 f1=0.270
 | lr=1.00e-04
  Epoch 022 | train loss=1.0701 acc=0.397 f1=0.388 | val loss=1.1151 acc=0.322 f1=0.296
 | lr=1.00e-04
  Epoch 023 | train loss=1.0702 acc=0.384 f1=0.371 | val loss=1.1142 acc=0.336 f1=0.303
 | lr=1.00e-04
  Epoch 024 | train loss=1.0670 acc=0.383 f1=0.379 | val loss=1.1200 acc=0.329 f1=0.303
 | lr=1.00e-04
  Epoch 025 | train loss=1.0676 acc=0.390 f1=0.383 | val loss=1.1125 acc=0.333 f1=0.303
 | lr=1.00e-04
  Epoch 026 | train loss=1.0692 acc=0.394 f1=0.388 | val loss=1.1159 acc=0.316 f1=0.281
 | lr=5.00e-05
  Epoch 027 | train loss=1.0653 acc=0.400 f1=0.392 | val loss=1.1170 acc=0.331 f1=0.300
 | lr=5.00e-05
  Epoch 028 | train loss=1.0650 acc=0.407 f1=0.401 | val loss=1.1178 acc=0.331 f1=0.302
 | lr=5.00e-05
  Epoch 029 | train loss=1.0628 acc=0.400 f1=0.394 | val loss=1.1161 acc=0.298 f1=0.253
 | lr=5.00e-05
  Epoch 030 | train loss=1.0658 acc=0.407 f1=0.400 | val loss=1.1210 acc=0.307 f1=0.278
 | lr=5.00e-05
```

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File - eeg_cnn_lstm

```
Epoch 031 | train loss=1.0639 acc=0.407 f1=0.402 | val loss=1.1201 acc=0.324 f1=0.295
| lr=5.00e-05
Epoch 032 | train loss=1.0604 acc=0.409 f1=0.405 | val loss=1.1177 acc=0.336 f1=0.304
| lr=5.00e-05
Epoch 033 | train loss=1.0578 acc=0.415 f1=0.407 | val loss=1.1160 acc=0.329 f1=0.292
| lr=5.00e-05
Epoch 034 | train loss=1.0607 acc=0.400 f1=0.396 | val loss=1.1177 acc=0.313 f1=0.274
| lr=5.00e-05
Epoch 035 | train loss=1.0635 acc=0.396 f1=0.392 | val loss=1.1189 acc=0.340 f1=0.313
| lr=2.50e-05
Epoch 036 | train loss=1.0579 acc=0.409 f1=0.405 | val loss=1.1169 acc=0.336 f1=0.311
| lr=2.50e-05
Epoch 037 | train loss=1.0551 acc=0.423 f1=0.420 | val loss=1.1190 acc=0.307 f1=0.269
| lr=2.50e-05
Epoch 038 | train loss=1.0571 acc=0.411 f1=0.405 | val loss=1.1201 acc=0.336 f1=0.309
| lr=2.50e-05
Epoch 039 | train loss=1.0572 acc=0.400 f1=0.396 | val loss=1.1214 acc=0.333 f1=0.306
| lr=2.50e-05
Epoch 040 | train loss=1.0555 acc=0.412 f1=0.408 | val loss=1.1220 acc=0.342 f1=0.314
| lr=2.50e-05
Epoch 041 | train loss=1.0604 acc=0.403 f1=0.398 | val loss=1.1230 acc=0.324 f1=0.291
| lr=2.50e-05
Epoch 042 | train loss=1.0585 acc=0.424 f1=0.418 | val loss=1.1197 acc=0.331 f1=0.303
| lr=2.50e-05
Epoch 043 | train loss=1.0562 acc=0.417 f1=0.412 | val loss=1.1221 acc=0.318 f1=0.293
| lr=2.50e-05
Epoch 044 | train loss=1.0554 acc=0.420 f1=0.414 | val loss=1.1219 acc=0.331 f1=0.312
| lr=2.50e-05
Epoch 045 | train loss=1.0519 acc=0.420 f1=0.416 | val loss=1.1249 acc=0.329 f1=0.299
| lr=2.50e-05
Epoch 046 | train loss=1.0511 acc=0.424 f1=0.418 | val loss=1.1216 acc=0.320 f1=0.281
| lr=2.50e-05
Epoch 047 | train loss=1.0543 acc=0.416 f1=0.411 | val loss=1.1200 acc=0.307 f1=0.252
| lr=2.50e-05
Epoch 048 | train loss=1.0563 acc=0.412 f1=0.410 | val loss=1.1210 acc=0.302 f1=0.248
| lr=2.50e-05
Epoch 049 | train loss=1.0555 acc=0.414 f1=0.411 | val loss=1.1259 acc=0.336 f1=0.307
| lr=1.25e-05
Epoch 050 | train loss=1.0538 acc=0.409 f1=0.405 | val loss=1.1268 acc=0.316 f1=0.277
| lr=1.25e-05
Epoch 051 | train loss=1.0546 acc=0.417 f1=0.413 | val loss=1.1249 acc=0.311 f1=0.286
| lr=1.25e-05
Epoch 052 | train loss=1.0518 acc=0.427 f1=0.423 | val loss=1.1226 acc=0.311 f1=0.272
| lr=1.25e-05
Epoch 053 | train loss=1.0506 acc=0.422 f1=0.419 | val loss=1.1246 acc=0.327 f1=0.295
| lr=1.25e-05
Epoch 054 | train loss=1.0520 acc=0.433 f1=0.428 | val loss=1.1250 acc=0.318 f1=0.293
| lr=1.25e-05
Epoch 055 | train loss=1.0571 acc=0.411 f1=0.409 | val loss=1.1255 acc=0.316 f1=0.284
| lr=1.25e-05
Epoch 056 | train loss=1.0525 acc=0.424 f1=0.418 | val loss=1.1266 acc=0.327 f1=0.306
| lr=1.25e-05
Epoch 057 | train loss=1.0496 acc=0.418 f1=0.415 | val loss=1.1287 acc=0.331 f1=0.307
| lr=1.25e-05
Epoch 058 | train loss=1.0500 acc=0.416 f1=0.413 | val loss=1.1238 acc=0.311 f1=0.285
| lr=6.25e-06
Epoch 059 | train loss=1.0550 acc=0.414 f1=0.409 | val loss=1.1257 acc=0.322 f1=0.300
| lr=6.25e-06
Epoch 060 | train loss=1.0548 acc=0.425 f1=0.420 | val loss=1.1254 acc=0.316 f1=0.283
| lr=6.25e-06
Epoch 061 | train loss=1.0534 acc=0.426 f1=0.422 | val loss=1.1267 acc=0.313 f1=0.285
| lr=6.25e-06
Epoch 062 | train loss=1.0532 acc=0.422 f1=0.419 | val loss=1.1266 acc=0.318 f1=0.290
| lr=6.25e-06
Epoch 063 | train loss=1.0513 acc=0.422 f1=0.417 | val loss=1.1272 acc=0.311 f1=0.278
| lr=6.25e-06
Epoch 064 | train loss=1.0522 acc=0.409 f1=0.405 | val loss=1.1259 acc=0.311 f1=0.284
| lr=6.25e-06
Epoch 065 | train loss=1.0553 acc=0.407 f1=0.402 | val loss=1.1223 acc=0.304 f1=0.257
| lr=6.25e-06
Epoch 066 | train loss=1.0528 acc=0.420 f1=0.416 | val loss=1.1242 acc=0.316 f1=0.286
| lr=6.25e-06
Epoch 067 | train loss=1.0522 acc=0.414 f1=0.410 | val loss=1.1238 acc=0.313 f1=0.271
| lr=3.13e-06
Epoch 068 | train loss=1.0495 acc=0.418 f1=0.413 | val loss=1.1251 acc=0.318 f1=0.288
| lr=3.13e-06
Epoch 069 | train loss=1.0517 acc=0.425 f1=0.420 | val loss=1.1279 acc=0.316 f1=0.277
| lr=3.13e-06
```

File - eeg_cnn_lstm

```
  Epoch 070 | train loss=1.0512 acc=0.432 f1=0.429 | val loss=1.1289 acc=0.329 f1=0.307
 | lr=3.13e-06

  Early stopping at epoch 70. Best epoch: 40

-----------------------------------------------------------------
  TEST RESULTS
  Accuracy    : 0.3521  (35.2%)
  F1 macro    : 0.3368
  Per class   : Task1=0.221  Task2=0.394  Task3=0.395
  Precision   : Task1=0.341  Task2=0.360  Task3=0.349
  Recall      : Task1=0.164  Task2=0.435  Task3=0.456
  Best epoch : 40
  Confusion matrix:
    [29, 67, 81]  ← Task 1 predicted as
    [28, 77, 72]  ← Task 2 predicted as
    [28, 70, 82]  ← Task 3 predicted as
-----------------------------------------------------------------

  Model   → ./results/eeg_cnn_lstm/eeg_cnn_lstm.pt
  History → ./results/eeg_cnn_lstm/eeg_cnn_lstm_history.npy
  Summary → ./results/eeg_cnn_lstm/eeg_cnn_lstm_summary.json

Process finished with exit code 0
```

B.4 EMG CNN LSTM Results

File - emg_cnn_lstm

```
/Users/muskaan_garg_/GitHub/eeg_emg_fusion_ml/.venv/bin/python /Users/muskaan_garg_/
GitHub/eeg_emg_fusion_ml/models_sub_independant/emg_cnn_lstm.py
-----------------------------------------------------------------
  EMG HYBRID CNN-LSTM — Subject-Independent NeBULA
  Raw signal branch + handcrafted feature branch
-----------------------------------------------------------------
  Device         : mps
  Raw input      : (4292, 11, 240)
  Feature input  : (4292, 132)
  Kept starts    : [120, 160, 200, 240, 280, 320]
  Train subjects : [1, 2, 4, 5, 6, 7, 10, 11, 14, 15, 16, 18, 19, 20, 23, 25, 29, 31, 34
, 35, 36, 37, 38, 39, 40]
  Val subjects   : [13, 22, 26, 27, 32]
  Test subjects  : [8, 9, 17, 24, 30, 33]
  Parameters     : 105,700

  train:  2980 windows  (Task0=980, Task1=1000, Task2=1000)
  val  :   600 windows  (Task0=200, Task1=200, Task2=200)
  test :   712 windows  (Task0=236, Task1=236, Task2=240)

  Epoch 001 | train loss=0.7639 acc=0.700 f1=0.682 | val loss=0.5874 acc=0.793 f1=0.783
 | lr=5.00e-04
  Epoch 002 | train loss=0.5489 acc=0.804 f1=0.804 | val loss=0.4693 acc=0.858 f1=0.854
 | lr=5.00e-04
  Epoch 003 | train loss=0.4912 acc=0.834 f1=0.835 | val loss=0.4529 acc=0.868 f1=0.865
 | lr=5.00e-04
  Epoch 004 | train loss=0.4720 acc=0.839 f1=0.839 | val loss=0.4441 acc=0.870 f1=0.868
 | lr=5.00e-04
  Epoch 005 | train loss=0.4361 acc=0.859 f1=0.860 | val loss=0.3974 acc=0.890 f1=0.888
 | lr=5.00e-04
  Epoch 006 | train loss=0.4357 acc=0.869 f1=0.870 | val loss=0.4776 acc=0.872 f1=0.868
 | lr=5.00e-04
  Epoch 007 | train loss=0.4350 acc=0.867 f1=0.868 | val loss=0.4405 acc=0.878 f1=0.876
 | lr=5.00e-04
  Epoch 008 | train loss=0.3978 acc=0.887 f1=0.888 | val loss=0.4441 acc=0.887 f1=0.883
 | lr=5.00e-04
  Epoch 009 | train loss=0.3861 acc=0.899 f1=0.900 | val loss=0.4447 acc=0.872 f1=0.867
 | lr=5.00e-04
  Epoch 010 | train loss=0.3666 acc=0.905 f1=0.905 | val loss=0.4370 acc=0.875 f1=0.871
 | lr=5.00e-04
  Epoch 011 | train loss=0.3522 acc=0.915 f1=0.915 | val loss=0.4977 acc=0.845 f1=0.837
 | lr=5.00e-04
  Epoch 012 | train loss=0.3672 acc=0.907 f1=0.907 | val loss=0.4720 acc=0.862 f1=0.857
 | lr=5.00e-04
  Epoch 013 | train loss=0.3327 acc=0.927 f1=0.927 | val loss=0.4131 acc=0.890 f1=0.890
 | lr=5.00e-04
  Epoch 014 | train loss=0.3225 acc=0.929 f1=0.929 | val loss=0.4339 acc=0.885 f1=0.882
 | lr=5.00e-04
  Epoch 015 | train loss=0.3287 acc=0.928 f1=0.928 | val loss=0.4855 acc=0.853 f1=0.849
 | lr=5.00e-04
  Epoch 016 | train loss=0.2973 acc=0.946 f1=0.946 | val loss=0.4353 acc=0.875 f1=0.873
 | lr=5.00e-04
  Epoch 017 | train loss=0.3051 acc=0.939 f1=0.939 | val loss=0.4320 acc=0.877 f1=0.874
 | lr=5.00e-04
  Epoch 018 | train loss=0.3029 acc=0.942 f1=0.942 | val loss=0.4439 acc=0.870 f1=0.867
 | lr=5.00e-04
  Epoch 019 | train loss=0.2789 acc=0.957 f1=0.957 | val loss=0.4638 acc=0.867 f1=0.864
 | lr=5.00e-04
  Epoch 020 | train loss=0.2859 acc=0.950 f1=0.950 | val loss=0.4760 acc=0.845 f1=0.840
 | lr=5.00e-04
  Epoch 021 | train loss=0.2807 acc=0.955 f1=0.955 | val loss=0.4605 acc=0.867 f1=0.862
 | lr=5.00e-04
  Epoch 022 | train loss=0.2652 acc=0.964 f1=0.964 | val loss=0.4949 acc=0.855 f1=0.849
 | lr=2.50e-04
  Epoch 023 | train loss=0.2662 acc=0.960 f1=0.960 | val loss=0.4853 acc=0.858 f1=0.853
 | lr=2.50e-04
  Epoch 024 | train loss=0.2536 acc=0.965 f1=0.966 | val loss=0.4573 acc=0.873 f1=0.871
 | lr=2.50e-04
  Epoch 025 | train loss=0.2441 acc=0.972 f1=0.973 | val loss=0.4584 acc=0.865 f1=0.862
 | lr=2.50e-04
  Epoch 026 | train loss=0.2435 acc=0.973 f1=0.973 | val loss=0.4986 acc=0.850 f1=0.845
 | lr=2.50e-04
  Epoch 027 | train loss=0.2448 acc=0.972 f1=0.973 | val loss=0.4623 acc=0.860 f1=0.856
 | lr=2.50e-04
  Epoch 028 | train loss=0.2346 acc=0.979 f1=0.979 | val loss=0.4632 acc=0.863 f1=0.860
 | lr=2.50e-04
  Epoch 029 | train loss=0.2356 acc=0.978 f1=0.978 | val loss=0.4514 acc=0.867 f1=0.864
 | lr=2.50e-04
```

File - emg_cnn_lstm

```
  Epoch 030 | train loss=0.2351 acc=0.977 f1=0.977 | val loss=0.4813 acc=0.845 f1=0.840
 | lr=2.50e-04
  Epoch 031 | train loss=0.2314 acc=0.980 f1=0.980 | val loss=0.4653 acc=0.862 f1=0.858
 | lr=1.25e-04
  Epoch 032 | train loss=0.2220 acc=0.987 f1=0.987 | val loss=0.4609 acc=0.862 f1=0.859
 | lr=1.25e-04
  Epoch 033 | train loss=0.2232 acc=0.984 f1=0.984 | val loss=0.4579 acc=0.872 f1=0.868
 | lr=1.25e-04
  Epoch 034 | train loss=0.2259 acc=0.983 f1=0.983 | val loss=0.4526 acc=0.868 f1=0.866
 | lr=1.25e-04
  Epoch 035 | train loss=0.2286 acc=0.982 f1=0.982 | val loss=0.4857 acc=0.852 f1=0.848
 | lr=1.25e-04
  Epoch 036 | train loss=0.2194 acc=0.987 f1=0.987 | val loss=0.4537 acc=0.868 f1=0.866
 | lr=1.25e-04
  Epoch 037 | train loss=0.2190 acc=0.988 f1=0.988 | val loss=0.4985 acc=0.848 f1=0.842
 | lr=1.25e-04
  Epoch 038 | train loss=0.2218 acc=0.988 f1=0.988 | val loss=0.4737 acc=0.858 f1=0.855
 | lr=1.25e-04
  Epoch 039 | train loss=0.2194 acc=0.986 f1=0.986 | val loss=0.4637 acc=0.863 f1=0.860
 | lr=1.25e-04
  Epoch 040 | train loss=0.2196 acc=0.986 f1=0.986 | val loss=0.4685 acc=0.852 f1=0.849
 | lr=6.25e-05
  Epoch 041 | train loss=0.2182 acc=0.987 f1=0.987 | val loss=0.4618 acc=0.865 f1=0.862
 | lr=6.25e-05
  Epoch 042 | train loss=0.2144 acc=0.989 f1=0.989 | val loss=0.4792 acc=0.857 f1=0.854
 | lr=6.25e-05
  Epoch 043 | train loss=0.2193 acc=0.984 f1=0.984 | val loss=0.4780 acc=0.860 f1=0.857
 | lr=6.25e-05

  Early stopping at epoch 43. Best epoch: 13

-----------------------------------------------------------------
  TEST RESULTS
  Accuracy   : 0.8413  (84.1%)
  F1 macro   : 0.8422
  Per class  : Task1=0.846  Task2=0.807  Task3=0.874
  Precision  : Task1=0.943  Task2=0.747  Task3=0.868
  Recall     : Task1=0.767  Task2=0.877  Task3=0.879
  Best epoch : 13
  Confusion matrix:
    [181, 41, 14]  ← Task 1 predicted as
    [11, 207, 18]  ← Task 2 predicted as
    [0, 29, 211]  ← Task 3 predicted as
-----------------------------------------------------------------

  Model   → ./results/emg_cnn_lstm/emg_hybrid_cnn_lstm.pt
  History → ./results/emg_cnn_lstm/emg_hybrid_cnn_lstm_history.npy
  Summary → ./results/emg_cnn_lstm/emg_hybrid_cnn_lstm_summary.json

Process finished with exit code 0
```

B.5 Fusion Results

File - fusion

```
/Users/muskaan_garg_/GitHub/eeg_emg_fusion_ml/.venv/bin/python /Users/muskaan_garg_/
GitHub/eeg_emg_fusion_ml/models_sub_independant/fusion.py
-----------------------------------------------------------------
  EEG-EMG Fusion — Subject-Independent NeBULA Classification
  EMG-anchored with learned EEG gate
-----------------------------------------------------------------
  Device         : mps
  EMG raw input  : (4292, 11, 240)
  EMG feat input : (4292, 44)
  EEG raw input  : (4292, 15, 240)
  EMG windows    : [120, 160, 200, 240, 280, 320]
  EEG windows    : [40, 80, 120]
  Train subjects : [1, 2, 4, 5, 6, 7, 10, 11, 14, 15, 16, 18, 19, 20, 23, 25, 29, 31, 34
, 35, 36, 37, 38, 39, 40]
  Val subjects   : [13, 22, 26, 27, 32]
  Test subjects  : [8, 9, 17, 24, 30, 33]
  Parameters     : 158,952

  train:  2980 samples  (Task0=980, Task1=1000, Task2=1000)
  val  :   600 samples  (Task0=200, Task1=200, Task2=200)
  test :   712 samples  (Task0=236, Task1=236, Task2=240)

  Epoch 001 | train loss=0.7773 acc=0.703 f1=0.690 | val loss=0.5664 acc=0.790 f1=0.780
 | lr=5.00e-04
  Epoch 002 | train loss=0.5275 acc=0.815 f1=0.815 | val loss=0.4987 acc=0.843 f1=0.839
 | lr=5.00e-04
  Epoch 003 | train loss=0.4544 acc=0.860 f1=0.860 | val loss=0.4623 acc=0.855 f1=0.852
 | lr=5.00e-04
  Epoch 004 | train loss=0.3980 acc=0.897 f1=0.897 | val loss=0.4786 acc=0.867 f1=0.866
 | lr=5.00e-04
  Epoch 005 | train loss=0.3690 acc=0.907 f1=0.907 | val loss=0.4914 acc=0.863 f1=0.859
 | lr=5.00e-04
  Epoch 006 | train loss=0.3466 acc=0.921 f1=0.921 | val loss=0.4989 acc=0.850 f1=0.847
 | lr=5.00e-04
  Epoch 007 | train loss=0.3164 acc=0.937 f1=0.937 | val loss=0.4721 acc=0.877 f1=0.876
 | lr=5.00e-04
  Epoch 008 | train loss=0.3058 acc=0.943 f1=0.943 | val loss=0.5607 acc=0.833 f1=0.827
 | lr=5.00e-04
  Epoch 009 | train loss=0.2891 acc=0.953 f1=0.953 | val loss=0.5609 acc=0.842 f1=0.834
 | lr=5.00e-04
  Epoch 010 | train loss=0.2683 acc=0.963 f1=0.963 | val loss=0.5039 acc=0.877 f1=0.873
 | lr=5.00e-04
  Epoch 011 | train loss=0.2533 acc=0.973 f1=0.974 | val loss=0.5213 acc=0.857 f1=0.854
 | lr=5.00e-04
  Epoch 012 | train loss=0.2589 acc=0.972 f1=0.972 | val loss=0.5330 acc=0.863 f1=0.860
 | lr=5.00e-04
  Epoch 013 | train loss=0.2382 acc=0.980 f1=0.980 | val loss=0.5216 acc=0.863 f1=0.860
 | lr=5.00e-04
  Epoch 014 | train loss=0.2395 acc=0.980 f1=0.980 | val loss=0.4987 acc=0.862 f1=0.859
 | lr=5.00e-04
  Epoch 015 | train loss=0.2316 acc=0.984 f1=0.984 | val loss=0.5244 acc=0.865 f1=0.862
 | lr=5.00e-04
  Epoch 016 | train loss=0.2390 acc=0.980 f1=0.980 | val loss=0.5232 acc=0.872 f1=0.871
 | lr=2.50e-04
  Epoch 017 | train loss=0.2252 acc=0.988 f1=0.988 | val loss=0.5173 acc=0.862 f1=0.860
 | lr=2.50e-04
  Epoch 018 | train loss=0.2184 acc=0.989 f1=0.989 | val loss=0.5436 acc=0.852 f1=0.850
 | lr=2.50e-04
  Epoch 019 | train loss=0.2154 acc=0.992 f1=0.992 | val loss=0.5921 acc=0.855 f1=0.852
 | lr=2.50e-04
  Epoch 020 | train loss=0.2205 acc=0.987 f1=0.987 | val loss=0.5945 acc=0.838 f1=0.834
 | lr=2.50e-04
  Epoch 021 | train loss=0.2146 acc=0.990 f1=0.990 | val loss=0.5861 acc=0.847 f1=0.843
 | lr=2.50e-04
  Epoch 022 | train loss=0.2103 acc=0.994 f1=0.994 | val loss=0.6028 acc=0.830 f1=0.824
 | lr=2.50e-04
  Epoch 023 | train loss=0.2112 acc=0.993 f1=0.993 | val loss=0.5968 acc=0.838 f1=0.834
 | lr=2.50e-04
  Epoch 024 | train loss=0.2080 acc=0.992 f1=0.992 | val loss=0.5534 acc=0.857 f1=0.854
 | lr=2.50e-04
  Epoch 025 | train loss=0.2107 acc=0.993 f1=0.993 | val loss=0.5702 acc=0.857 f1=0.853
 | lr=1.25e-04
  Epoch 026 | train loss=0.2065 acc=0.995 f1=0.995 | val loss=0.5834 acc=0.845 f1=0.841
 | lr=1.25e-04
  Epoch 027 | train loss=0.2075 acc=0.992 f1=0.992 | val loss=0.5607 acc=0.858 f1=0.855
 | lr=1.25e-04
  Epoch 028 | train loss=0.2060 acc=0.992 f1=0.992 | val loss=0.5732 acc=0.852 f1=0.848
 | lr=1.25e-04
```

File - fusion

```
  Epoch 029 | train loss=0.2045 acc=0.995 f1=0.995 | val loss=0.5748 acc=0.852 f1=0.848
 | lr=1.25e-04
  Epoch 030 | train loss=0.2067 acc=0.993 f1=0.993 | val loss=0.5806 acc=0.855 f1=0.853
 | lr=1.25e-04
  Epoch 031 | train loss=0.1990 acc=0.999 f1=0.999 | val loss=0.5881 acc=0.843 f1=0.841
 | lr=1.25e-04
  Epoch 032 | train loss=0.1996 acc=0.997 f1=0.997 | val loss=0.5837 acc=0.845 f1=0.842
 | lr=1.25e-04
  Epoch 033 | train loss=0.2016 acc=0.996 f1=0.996 | val loss=0.5742 acc=0.845 f1=0.841
 | lr=1.25e-04
  Epoch 034 | train loss=0.2016 acc=0.997 f1=0.997 | val loss=0.5799 acc=0.855 f1=0.852
 | lr=6.25e-05
  Epoch 035 | train loss=0.2020 acc=0.994 f1=0.994 | val loss=0.6157 acc=0.843 f1=0.839
 | lr=6.25e-05
  Epoch 036 | train loss=0.1963 acc=0.998 f1=0.998 | val loss=0.5697 acc=0.850 f1=0.847
 | lr=6.25e-05
  Epoch 037 | train loss=0.1996 acc=0.998 f1=0.998 | val loss=0.5678 acc=0.848 f1=0.845
 | lr=6.25e-05

  Early stopping at epoch 37. Best epoch: 7

--------------------------------------------------------
  TEST RESULTS
  Accuracy   : 0.7823  (78.2%)
  F1 macro   : 0.7840
  Per class  : Task1=0.800  Task2=0.737  Task3=0.815
  Precision  : Task1=0.943  Task2=0.677  Task3=0.789
  Recall     : Task1=0.695  Task2=0.809  Task3=0.842
  Best epoch : 7
  Confusion matrix:
    [164, 54, 18]  ← Task 1 predicted as
    [9, 191, 36]  ← Task 2 predicted as
    [1, 37, 202]  ← Task 3 predicted as
-----------------------------------------------------------------

  Model   → ./results/fusion/fusion.pt
  History → ./results/fusion/fusion_history.npy
  Summary → ./results/fusion/fusion_summary.json

Process finished with exit code 0
```

C. Subject Dependent Results

C.1 EEGNet Subject Dependant Results

File - eeg_net_dep

```
/Users/muskaan_garg_/GitHub/eeg_emg_fusion_ml/.venv/bin/python /Users/muskaan_garg_/
GitHub/eeg_emg_fusion_ml/subject_dependant/models_sub_dep/eeg_net_dep.py
------------------------------------------------------------
  EEGNet — Subject-Dependent NeBULA Classification
------------------------------------------------------------
  Subjects : [1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14, 15, 16, 17, 18, 19, 20, 22, 23, 24
, 25, 26, 27, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40]
  Config   : epochs=200, lr=0.0001, dropout=0.5, patience=40

  sub-01  acc=0.312  f1=0.159  (train=223win, test=48win, stopped@ep160)
  sub-02  acc=0.500  f1=0.278  (train=223win, test=48win, stopped@ep160)
  sub-04  acc=0.400  f1=0.310  (train=230win, test=50win, stopped@ep160)
  sub-05  acc=0.340  f1=0.169  (train=230win, test=50win, stopped@ep163)
  sub-06  acc=0.250  f1=0.160  (train=223win, test=48win, stopped@ep160)
  sub-07  acc=0.440  f1=0.204  (train=230win, test=50win, stopped@ep196)
  sub-08  acc=0.360  f1=0.203  (train=230win, test=50win, stopped@ep160)
  sub-09  acc=0.333  f1=0.193  (train=223win, test=48win, stopped@ep160)
  sub-10  acc=0.240  f1=0.159  (train=230win, test=50win, stopped@ep160)
  sub-11  acc=0.360  f1=0.273  (train=230win, test=50win, stopped@ep160)
  sub-13  acc=0.300  f1=0.261  (train=230win, test=50win, stopped@ep160)
  sub-14  acc=0.292  f1=0.153  (train=223win, test=48win, stopped@ep160)
  sub-15  acc=0.125  f1=0.123  (train=223win, test=48win, stopped@ep160)
  sub-16  acc=0.400  f1=0.190  (train=230win, test=50win, stopped@ep160)
  sub-17  acc=0.125  f1=0.109  (train=223win, test=48win, stopped@ep160)
  sub-18  acc=0.440  f1=0.204  (train=230win, test=50win, stopped@ep160)
  sub-19  acc=0.320  f1=0.162  (train=230win, test=50win, stopped@ep160)
  sub-20  acc=0.200  f1=0.111  (train=230win, test=50win, stopped@ep173)
  sub-22  acc=0.020  f1=0.021  (train=230win, test=50win, stopped@ep160)
  sub-23  acc=0.600  f1=0.346  (train=230win, test=50win, stopped@ep160)
  sub-24  acc=0.680  f1=0.364  (train=230win, test=50win, stopped@ep160)
  sub-25  acc=0.220  f1=0.124  (train=230win, test=50win, stopped@ep160)
  sub-26  acc=0.420  f1=0.203  (train=230win, test=50win, stopped@ep160)
  sub-27  acc=0.260  f1=0.167  (train=230win, test=50win, stopped@ep160)
  sub-29  acc=0.660  f1=0.265  (train=230win, test=50win, stopped@ep160)
  sub-30  acc=0.160  f1=0.105  (train=230win, test=50win, stopped@ep160)
  sub-31  acc=0.560  f1=0.367  (train=230win, test=50win, stopped@ep160)
  sub-32  acc=0.660  f1=0.265  (train=230win, test=50win, stopped@ep160)
  sub-33  acc=0.400  f1=0.239  (train=230win, test=50win, stopped@ep160)
  sub-34  acc=0.220  f1=0.122  (train=230win, test=50win, stopped@ep160)
  sub-35  acc=0.240  f1=0.223  (train=230win, test=50win, stopped@ep160)
  sub-36  acc=0.460  f1=0.295  (train=230win, test=50win, stopped@ep160)
  sub-37  acc=0.500  f1=0.335  (train=230win, test=50win, stopped@ep160)
  sub-38  acc=0.180  f1=0.164  (train=230win, test=50win, stopped@ep160)
  sub-39  acc=0.120  f1=0.075  (train=230win, test=50win, stopped@ep160)
  sub-40  acc=0.540  f1=0.319  (train=230win, test=50win, stopped@ep160)

------------------------------------------------------------
  Mean accuracy : 0.351 ± 0.164
  Mean F1       : 0.206  ± 0.084
  Best subject  : sub-24  (0.680)
  Worst subject : sub-22  (0.020)
------------------------------------------------------------

  Results saved → ../results_dep/eegnet_sub_dep/eegnet_sub_dep_results.npy

Process finished with exit code 0
```

Page 1 of 1

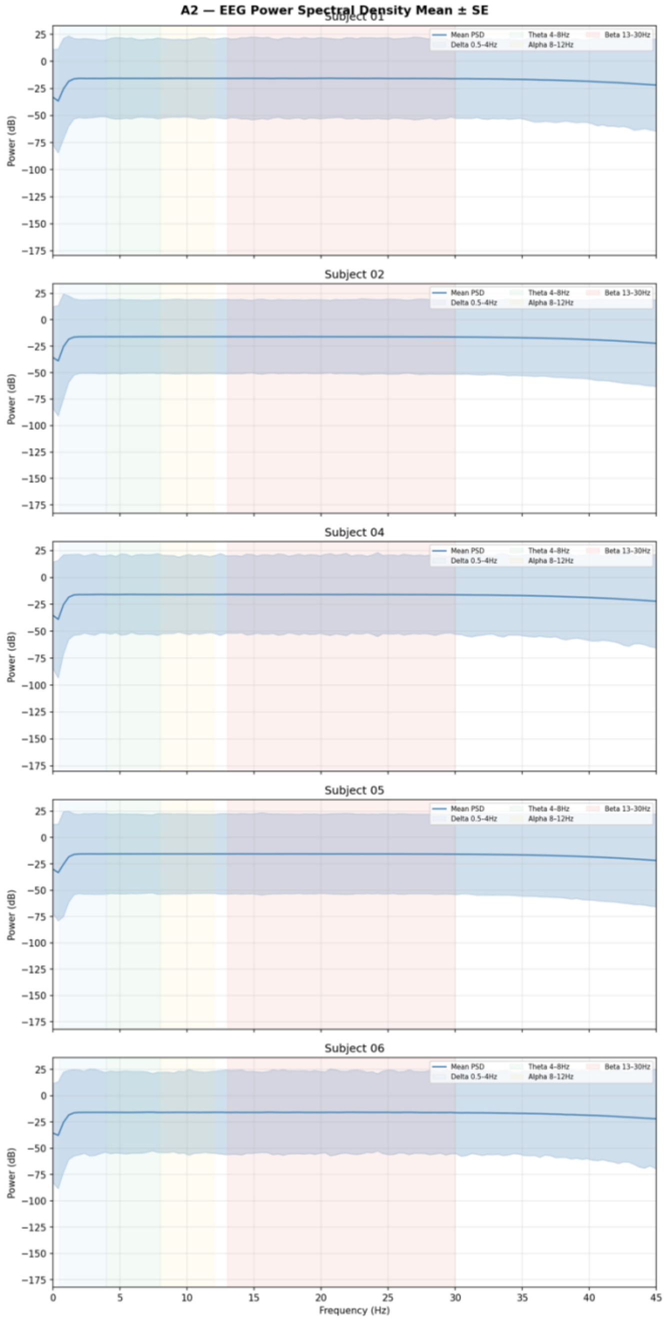
C.2 EMGNet Subject Dependant Results

File - emg_net_dep

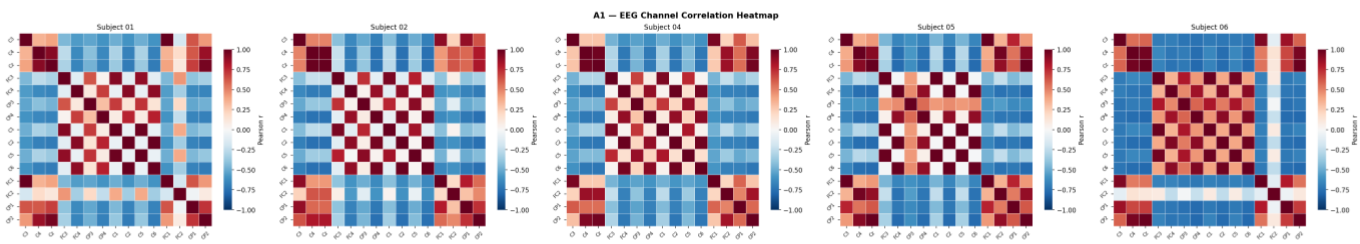
```
/Users/muskaan_garg_/GitHub/eeg_emg_fusion_ml/.venv/bin/python /Users/muskaan_garg_/
GitHub/eeg_emg_fusion_ml/subject_dependant/models_sub_dep/emg_net_dep.py
------------------------------------------------------------
  EMGNet — Subject-Dependent NeBULA Classification
------------------------------------------------------------
  Subjects : [1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14, 15, 16, 17, 18, 19, 20, 22, 23, 24
, 25, 26, 27, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40]
  Config   : epochs=200, lr=0.0001, dropout=0.25, patience=40

  sub-01  acc=0.562  f1=0.531  (train=223win, test=48win, stopped@ep160)
  sub-02  acc=0.688  f1=0.575  (train=223win, test=48win, stopped@ep160)
  sub-04  acc=0.300  f1=0.277  (train=230win, test=50win, stopped@ep160)
  sub-05  acc=0.720  f1=0.700  (train=230win, test=50win, stopped@ep160)
  sub-06  acc=0.688  f1=0.689  (train=223win, test=48win, stopped@ep160)
  sub-07  acc=0.660  f1=0.646  (train=230win, test=50win, stopped@ep160)
  sub-08  acc=0.420  f1=0.358  (train=230win, test=50win, stopped@ep160)
  sub-09  acc=0.833  f1=0.830  (train=223win, test=48win, stopped@ep160)
  sub-10  acc=0.640  f1=0.647  (train=230win, test=50win, stopped@ep160)
  sub-11  acc=0.680  f1=0.688  (train=230win, test=50win, stopped@ep160)
  sub-13  acc=0.620  f1=0.637  (train=230win, test=50win, stopped@ep160)
  sub-14  acc=0.500  f1=0.522  (train=223win, test=48win, stopped@ep160)
  sub-15  acc=0.583  f1=0.521  (train=223win, test=48win, stopped@ep160)
  sub-16  acc=0.700  f1=0.663  (train=230win, test=50win, stopped@ep160)
  sub-17  acc=0.583  f1=0.539  (train=223win, test=48win, stopped@ep160)
  sub-18  acc=0.760  f1=0.764  (train=230win, test=50win, stopped@ep160)
  sub-19  acc=0.780  f1=0.767  (train=230win, test=50win, stopped@ep160)
  sub-20  acc=0.680  f1=0.683  (train=230win, test=50win, stopped@ep160)
  sub-22  acc=0.640  f1=0.499  (train=230win, test=50win, stopped@ep160)
  sub-23  acc=0.760  f1=0.713  (train=230win, test=50win, stopped@ep160)
  sub-24  acc=0.780  f1=0.736  (train=230win, test=50win, stopped@ep160)
  sub-25  acc=0.700  f1=0.653  (train=230win, test=50win, stopped@ep160)
  sub-26  acc=0.760  f1=0.679  (train=230win, test=50win, stopped@ep160)
  sub-27  acc=0.200  f1=0.190  (train=230win, test=50win, stopped@ep160)
  sub-29  acc=0.720  f1=0.693  (train=230win, test=50win, stopped@ep160)
  sub-30  acc=0.740  f1=0.673  (train=230win, test=50win, stopped@ep160)
  sub-31  acc=0.480  f1=0.520  (train=230win, test=50win, stopped@ep160)
  sub-32  acc=0.640  f1=0.621  (train=230win, test=50win, stopped@ep160)
  sub-33  acc=0.760  f1=0.701  (train=230win, test=50win, stopped@ep160)
  sub-34  acc=0.640  f1=0.623  (train=230win, test=50win, stopped@ep160)
  sub-35  acc=0.760  f1=0.732  (train=230win, test=50win, stopped@ep160)
  sub-36  acc=0.800  f1=0.750  (train=230win, test=50win, stopped@ep160)
  sub-37  acc=0.440  f1=0.450  (train=230win, test=50win, stopped@ep160)
  sub-38  acc=0.600  f1=0.613  (train=230win, test=50win, stopped@ep160)
  sub-39  acc=0.740  f1=0.627  (train=230win, test=50win, stopped@ep160)
  sub-40  acc=0.760  f1=0.724  (train=230win, test=50win, stopped@ep160)

------------------------------------------------------------
  Mean accuracy : 0.648 ± 0.139
  Mean F1       : 0.618  ± 0.135
  Best subject  : sub-09  (0.833)
  Worst subject : sub-27  (0.200)
------------------------------------------------------------

  Results saved → ../results_dep/emgnet_sub_dep/emgnet_sub_dep_results.npy

Process finished with exit code 0
```

D. EEG Diagnostic Plots

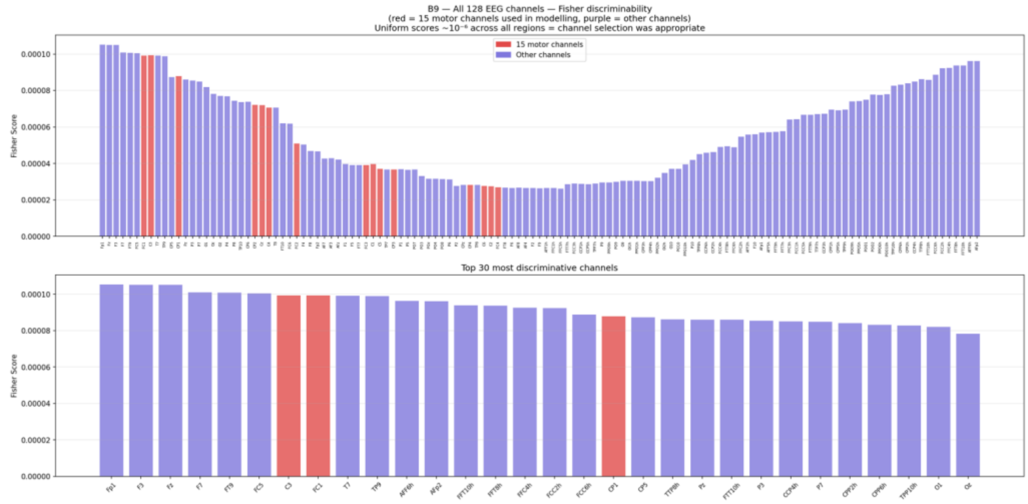
D.1 EEG PSD Mean SE



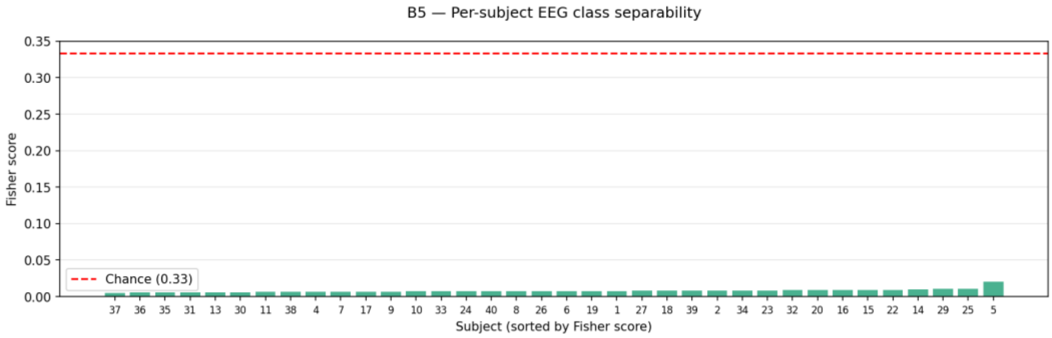
D.2 EEG Channel Correlation Heatmap



D.3 All EEG Channels – Fisher Discriminability

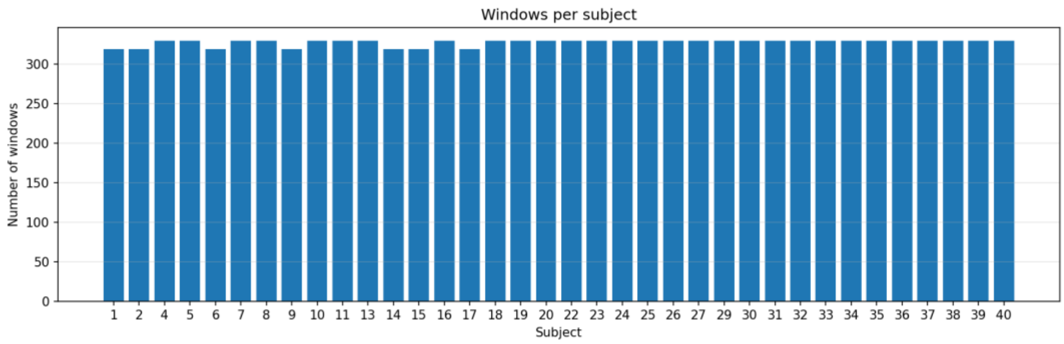


D.4 Per subject EEG separability

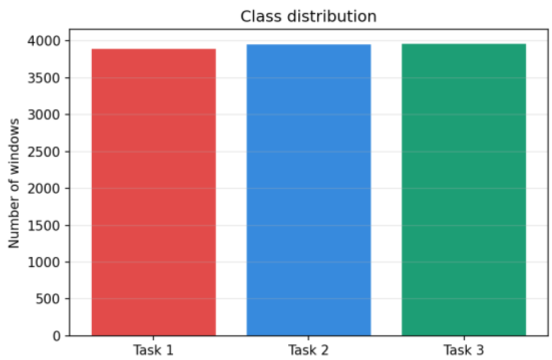


E. EMG Diagnostic Plots

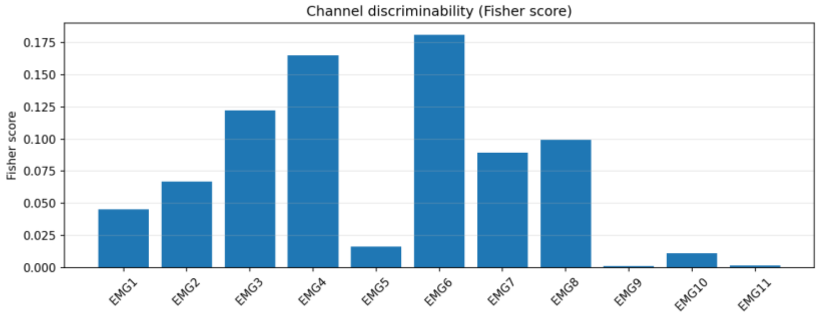
E.1 EMG windows per subject



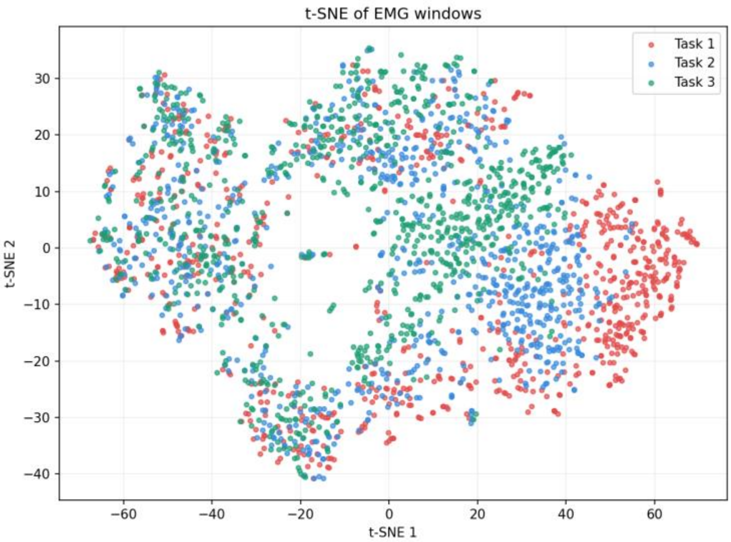
E.2 EMG class distribution



E.3 EMG class discriminability



E.4 EMG windows t-SNE



F. Temporal Pipeline Results

File - temporal_pipeline

```
/Users/muskaan_garg_/GitHub/eeg_emg_fusion_ml/.venv/bin/python /Users/muskaan_garg_/
GitHub/eeg_emg_fusion_ml/temporal_analysis/temporal_pipeline.py
-----------------------------------------------------------------
TIMING ANALYSIS PIPELINE
-----------------------------------------------------------------
Data dir              : ../preprocessed
Output dir            : ./timing_results
Conditions            : ['free']
EEG ROI               : ['C3', 'C4', 'Cz', 'FC3', 'FC4', 'CP3', 'CP4']
EMG threshold k       : 2.0
ERD drop fraction     : 0.1
EMG min duration ms   : 25
EEG min duration ms   : 30
EEG smoothing ms      : 50
-----------------------------------------------------------------

--- Subject 01 ---
  Condition 'free': EEG=(29, 15, 500), EMG=(29, 11, 500), y=(29,)

--- Subject 02 ---
  Condition 'free': EEG=(29, 15, 500), EMG=(29, 11, 500), y=(29,)

--- Subject 04 ---
  Condition 'free': EEG=(30, 15, 500), EMG=(30, 11, 500), y=(30,)

--- Subject 05 ---
  Condition 'free': EEG=(30, 15, 500), EMG=(30, 11, 500), y=(30,)

--- Subject 06 ---
  Condition 'free': EEG=(29, 15, 500), EMG=(29, 11, 500), y=(29,)

--- Subject 07 ---
  Condition 'free': EEG=(30, 15, 500), EMG=(30, 11, 500), y=(30,)

--- Subject 08 ---
  Condition 'free': EEG=(30, 15, 500), EMG=(30, 11, 500), y=(30,)

--- Subject 09 ---
  Condition 'free': EEG=(29, 15, 500), EMG=(29, 11, 500), y=(29,)

--- Subject 10 ---
  Condition 'free': EEG=(30, 15, 500), EMG=(30, 11, 500), y=(30,)

--- Subject 11 ---
  Condition 'free': EEG=(30, 15, 500), EMG=(30, 11, 500), y=(30,)

--- Subject 13 ---
  Condition 'free': EEG=(30, 15, 500), EMG=(30, 11, 500), y=(30,)

--- Subject 14 ---
  Condition 'free': EEG=(29, 15, 500), EMG=(29, 11, 500), y=(29,)

--- Subject 15 ---
  Condition 'free': EEG=(29, 15, 500), EMG=(29, 11, 500), y=(29,)

--- Subject 16 ---
  Condition 'free': EEG=(30, 15, 500), EMG=(30, 11, 500), y=(30,)

--- Subject 17 ---
  Condition 'free': EEG=(29, 15, 500), EMG=(29, 11, 500), y=(29,)

--- Subject 18 ---
  Condition 'free': EEG=(30, 15, 500), EMG=(30, 11, 500), y=(30,)

--- Subject 19 ---
  Condition 'free': EEG=(30, 15, 500), EMG=(30, 11, 500), y=(30,)

--- Subject 20 ---
  Condition 'free': EEG=(30, 15, 500), EMG=(30, 11, 500), y=(30,)

--- Subject 22 ---
  Condition 'free': EEG=(30, 15, 500), EMG=(30, 11, 500), y=(30,)

--- Subject 23 ---
  Condition 'free': EEG=(30, 15, 500), EMG=(30, 11, 500), y=(30,)

--- Subject 24 ---
  Condition 'free': EEG=(30, 15, 500), EMG=(30, 11, 500), y=(30,)
```

File - temporal_pipeline

```
--- Subject 25 ---
  Condition 'free': EEG=(30, 15, 500), EMG=(30, 11, 500), y=(30,)

--- Subject 26 ---
  Condition 'free': EEG=(30, 15, 500), EMG=(30, 11, 500), y=(30,)

--- Subject 27 ---
  Condition 'free': EEG=(30, 15, 500), EMG=(30, 11, 500), y=(30,)

--- Subject 29 ---
  Condition 'free': EEG=(30, 15, 500), EMG=(30, 11, 500), y=(30,)

--- Subject 30 ---
  Condition 'free': EEG=(30, 15, 500), EMG=(30, 11, 500), y=(30,)

--- Subject 31 ---
  Condition 'free': EEG=(30, 15, 500), EMG=(30, 11, 500), y=(30,)

--- Subject 32 ---
  Condition 'free': EEG=(30, 15, 500), EMG=(30, 11, 500), y=(30,)

--- Subject 33 ---
  Condition 'free': EEG=(30, 15, 500), EMG=(30, 11, 500), y=(30,)

--- Subject 34 ---
  Condition 'free': EEG=(30, 15, 500), EMG=(30, 11, 500), y=(30,)

--- Subject 35 ---
  Condition 'free': EEG=(30, 15, 500), EMG=(30, 11, 500), y=(30,)

--- Subject 36 ---
  Condition 'free': EEG=(30, 15, 500), EMG=(30, 11, 500), y=(30,)

--- Subject 37 ---
  Condition 'free': EEG=(30, 15, 500), EMG=(30, 11, 500), y=(30,)

--- Subject 38 ---
  Condition 'free': EEG=(30, 15, 500), EMG=(30, 11, 500), y=(30,)

--- Subject 39 ---
  Condition 'free': EEG=(30, 15, 500), EMG=(30, 11, 500), y=(30,)

--- Subject 40 ---
  Condition 'free': EEG=(30, 15, 500), EMG=(30, 11, 500), y=(30,)

------------------------------------------------------------------
CONDITION / TASK SUMMARY
------------------------------------------------------------------
    free | Task1 | n_sub=36 | delay= 179.22 ms | EEG=  -5.48 ms | EMG= 171.95 ms | XCorr
= -47.86 ms
    free | Task2 | n_sub=36 | delay= 181.08 ms | EEG= -23.40 ms | EMG= 159.05 ms | XCorr
=  10.55 ms
    free | Task3 | n_sub=36 | delay= 188.84 ms | EEG= -14.49 ms | EMG= 174.97 ms | XCorr
=   5.51 ms
Saved plot -> ./timing_results/01_delay_by_task_condition.png
Saved plot -> ./timing_results/02_trial_onset_scatter.png
Saved plot -> ./timing_results/03_delay_vs_xcorr.png

Done. Outputs saved in: ./timing_results

Process finished with exit code 0
```